Critical Thinking and Data Analysis



What is critical thinking?

- “Critical thinking is reasonable, reflective thinking that is focused on deciding what to believe and what to do" – Robert Ennis

- “Critical thinking is skillful, responsible thinking that is conductive to good judgement because it is sensitive to context, relies on criteria, and is self-correcting” – Matthew Lipman

- “Critical thinking is thinking about your thinking, while you’re thinking, in order to make your thinking better” – Richard Paul

Features of critical thinking

- Critical thinking is reflective: involves thinking about your thinking
- Critical thinking involves standards: creating criteria for judgement
- Critical thinking is authentic: involves thinking about real, “ill-posed” problems, where answers are not trivial
- Critical thinking is reasonable/thoughtful/mindful/flexible: sensitive to context, goals and other realities
- Critical thinking is not (necessarily) emotionless
- Critical thinking is not fast

Three parts of critical thinking

1. Critical thinking involves asking questions

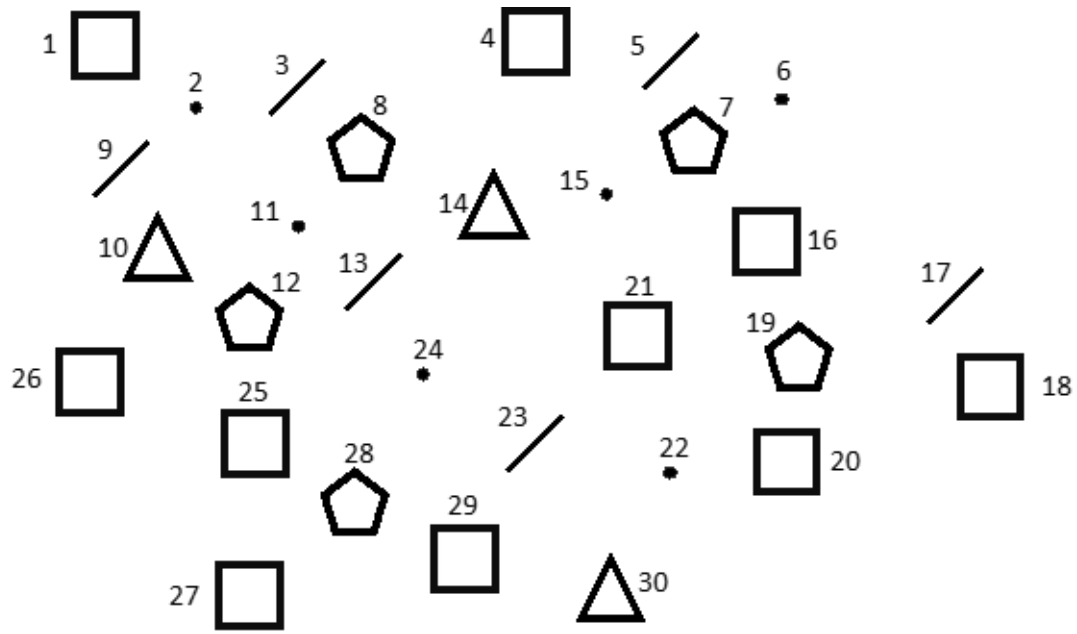
2. Critical thinking involves **trying** to answer those question by **reasoning** them out

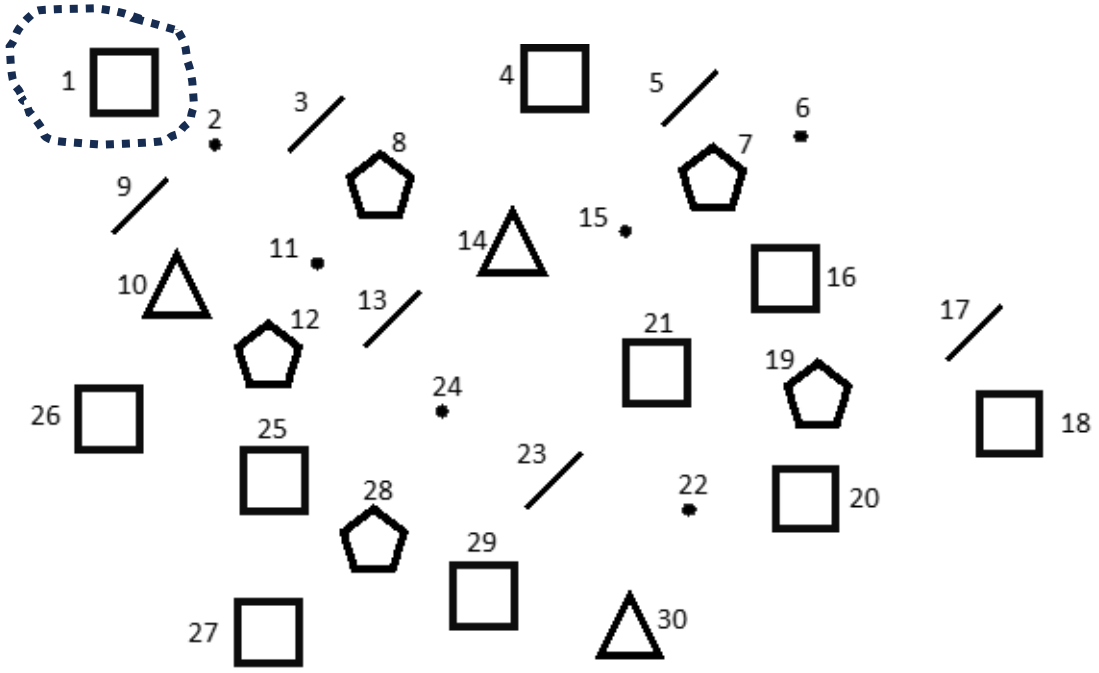
3. Critical thinking involves believing the result of our reasoning

“… guard against the idea that having no questions shows you understand something. For any subject matter that is at all complicated, the opposite is the case: If no questions arise for you, it probably means you do not understand it deeply enough to see the many paths of questioning that arise.”

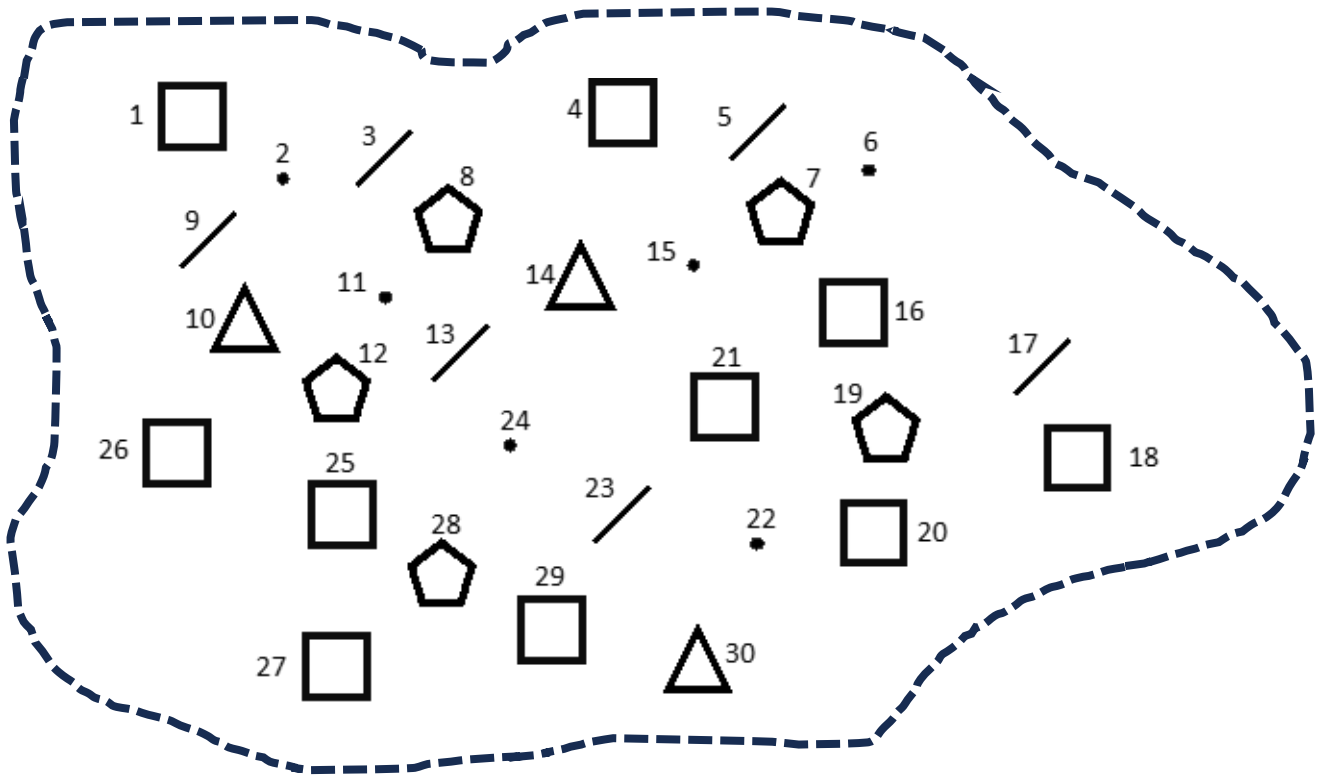
- Nosich (2012)

Some background on data

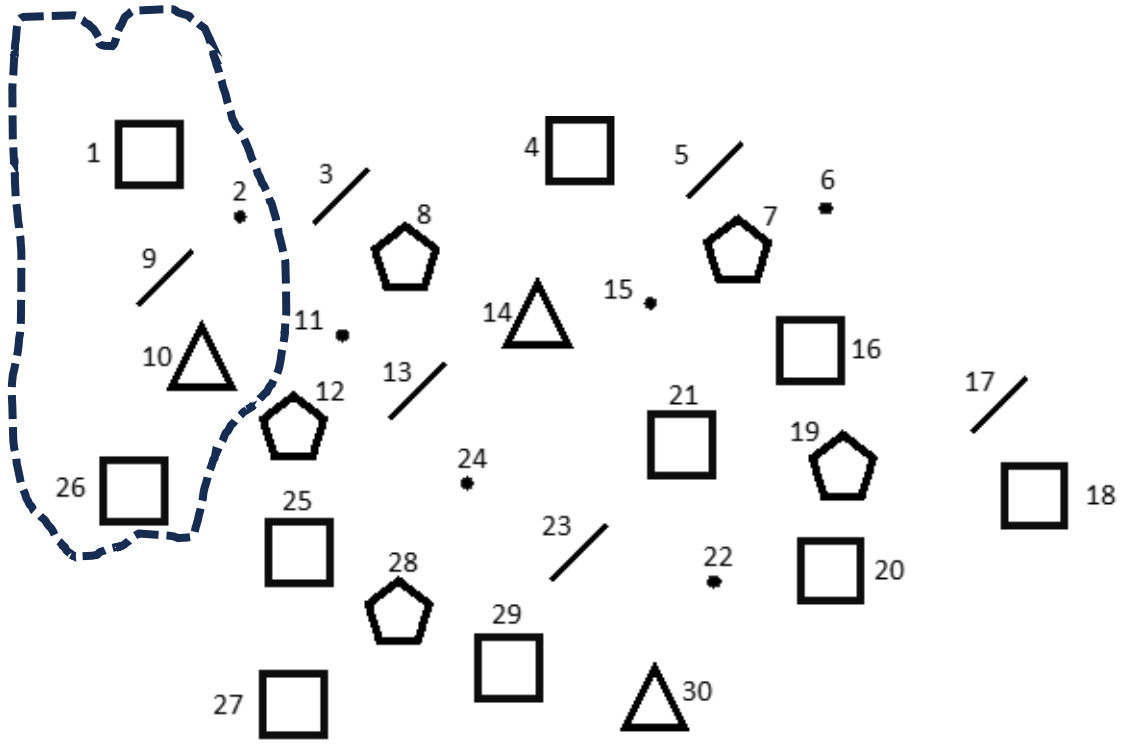

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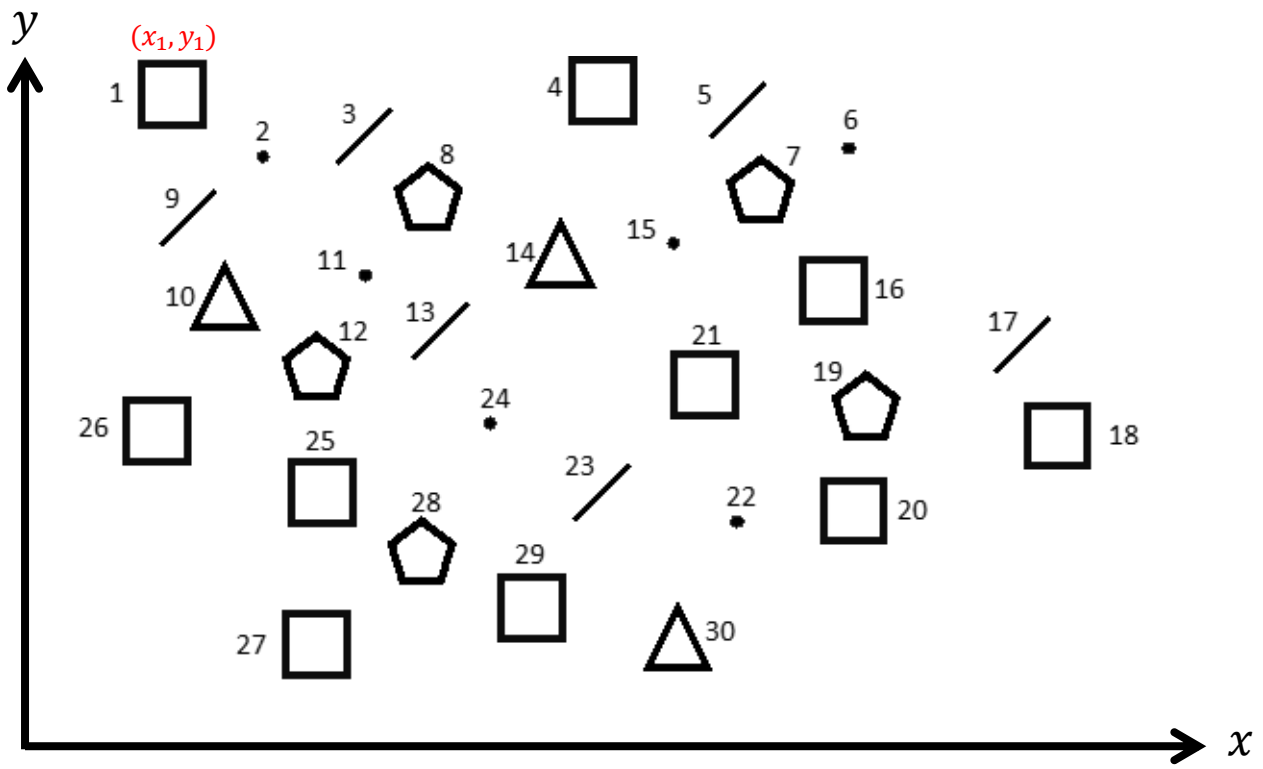
- **Object:** An individual unit of observation/measurement
  - E.g., one student, one playing card, one shape



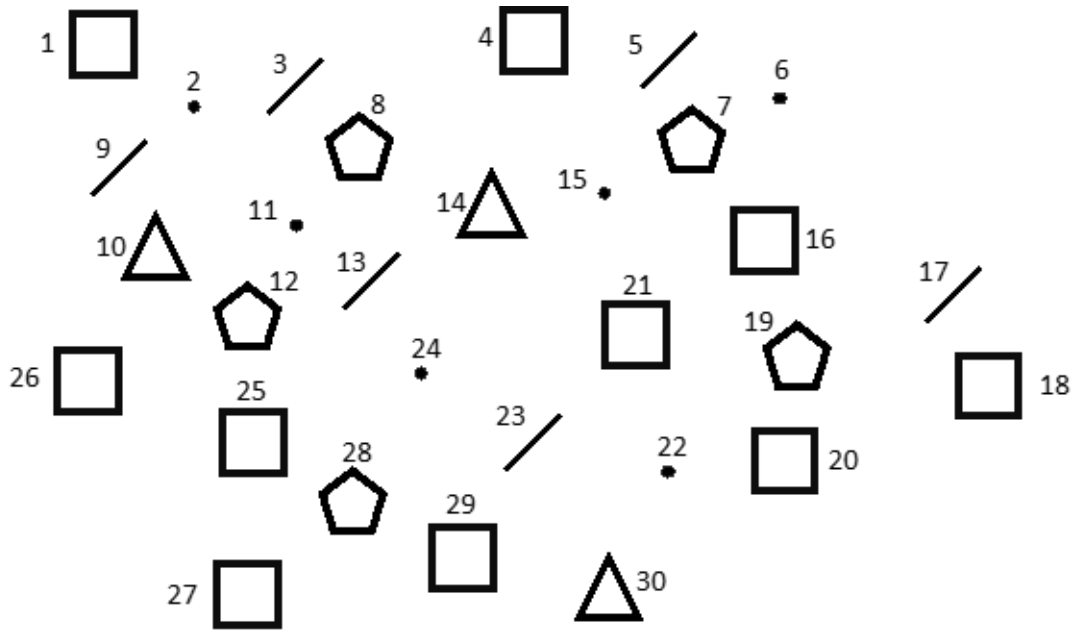
- **Population:** A set of objects (units of measurements)
  - E.g., a class of students, a deck of playing cards, a collection of shapes (1-30)



- **Sample:** A subset of objects (units of measurements)
  - E.g., one lab section of students, half a deck of playing cards, a subset of the shapes (1, 2, 9, 10, 26)



- **Property:** a descriptor/characteristic/measurement of an object
  - E.g., a student's GPA, a student's program, the suit of playing cards, the position of the shape



- **Constant:** a property that stays the same across **all** objects in the set
- **Variable:** a property that differs across some objects in the set

Datum – an individual piece of information

Data – multiple pieces of information; a collection of datum

Structured data – data that has *predictable* structure; can be readily analyzed

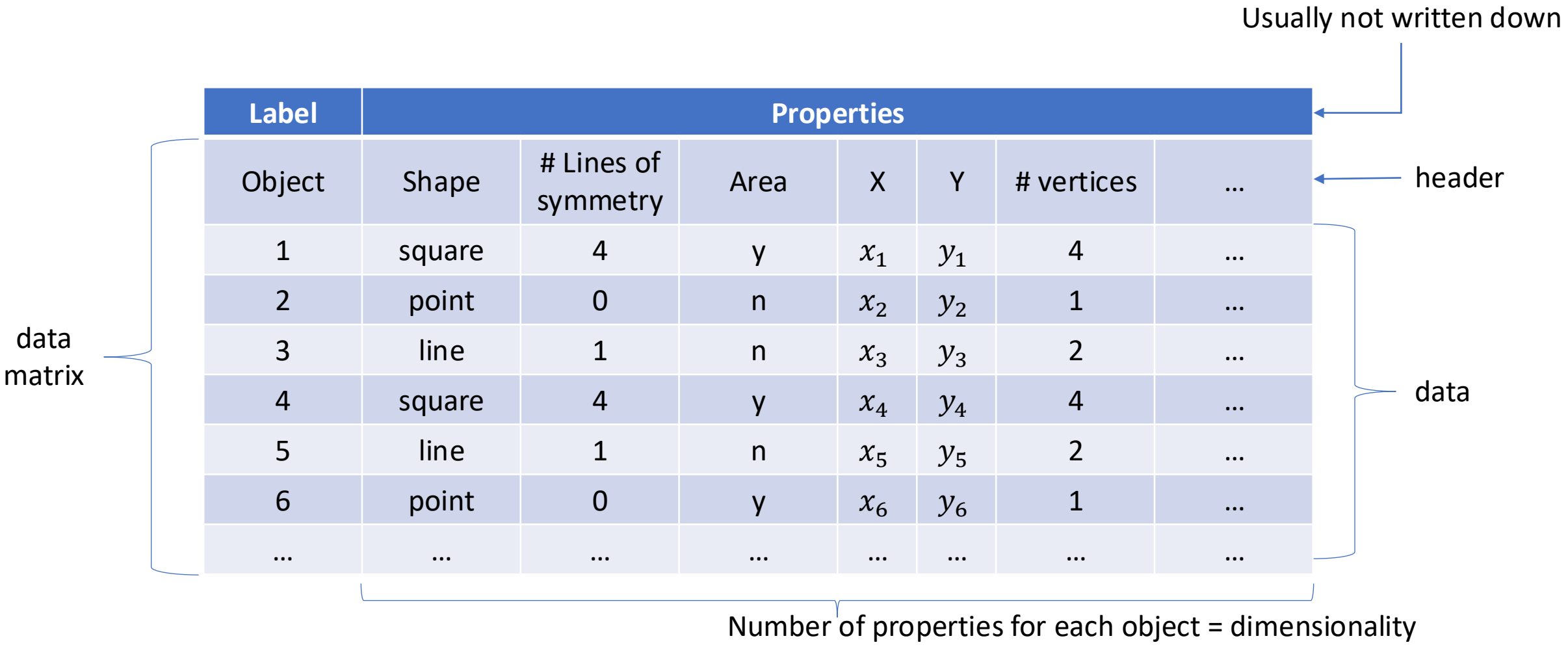
Unstructured data – data that has *unpredictable* structure; must be processed prior to analysis

Object 1 is a square; it has area, it is positioned at location (x_1, y_1) and has 4 vertices.

Object 2 is a point; it does not have area, it is positioned at location (x_2, y_2) and has 1 vertex.

. . .

Object <n> is a <insert shape>; it <insert area property>, it is positioned at location <insert coordinates> and has <number of vertices>.



- **Data matrix:** a standard format for recording and organizing the properties of objects in a sample/population (1 row of header information plus the data)

Dichotomous/binary variable

	Properties						
Object	Shape	# Lines of symmetry	Area	X	Y	# vertices	…
1	square	4	y	x_1	y_1	4	…
2	point	0	n	x_2	y_2	1	…
3	line	1	n	x_3	y_3	2	…
4	square	4	y	x_4	y_4	4	…
5	line	1	n	x_5	y_5	2	…
6	point	0	y	x_6	y_6	1	…
…	…	…	…	…	…	…	…

- **Nominal data:** data that can be be categorized in discrete categories where order of category has no significance

	Properties						
Object	Shape	# Lines of symmetry	Area	X	Y	# vertices	…
1	square	4	y	x_1	y_1	4	…
2	point	0	n	x_2	y_2	1	…
3	line	1	n	x_3	y_3	2	…
4	square	4	y	x_4	y_4	4	…
5	line	1	n	x_5	y_5	2	…
6	point	0	y	x_6	y_6	1	…
…	…	…	…	…	…	…	…

- **Ordinal data:** data that can be be categorized in discrete categories where order of category has significance

	Properties						
Object	Shape	# Lines of symmetry	Area	X	Y	# vertices	…
1	square	4	y	x_1	y_1	4	…
2	point	0	n	x_2	y_2	1	…
3	line	1	n	x_3	y_3	2	…
4	square	4	y	x_4	y_4	4	…
5	line	1	n	x_5	y_5	2	…
6	point	0	y	x_6	y_6	1	…
…	…	…	…	…	…	…	…

- **Continuous data:** data that can take on any value within a range

Case Study 1: Does shot gun brand effect pellet scatter pattern?

Target	gun	range	sqrt.area	Target	gun	range	sqrt.area	Target	gun	range	sqrt.area
1	Stevens	40	12.54	21	Stevens	20	6.96	41	Stevens	40	11.03
2	Remington	40	13.01	22	Stevens	20	6.32	42	Stevens	30	8.24
3	Remington	50	13.13	23	Stevens	40	11.47	43	Remington	20	5.73
4	Stevens	50	13.74	24	Remington	50	17.47	44	Stevens	30	7.38
5	Remington	10	3.29	25	Stevens	20	6.84	45	Stevens	10	2.6
6	Remington	30	7.95	26	Stevens	10	3.06	46	Remington	50	16.07
7	Remington	40	10.81	27	Stevens	10	3.35	47	Remington	50	14.9
8	Stevens	40	16.13	28	Remington	40	10.19	48	Stevens	20	6.29
9	Remington	10	3.01	29	Remington	20	5	49	Stevens	50	16.94
10	Remington	20	5.1	30	Remington	30	8.09	50	Stevens	50	14.09
11	Stevens	10	3.38	31	Remington	40	11.17	51	Remington	10	3.02
12	Remington	10	3.2	32	Remington	20	5.29	52	Stevens	20	5.85
13	Remington	30	8.62	33	Remington	50	14.21	53	Stevens	20	5.95
14	Remington	10	3.11	34	Remington	50	11.93	54	Stevens	50	20.13
15	Remington	40	11.33	35	Stevens	40	14.1	55	Stevens	10	3.33
16	Stevens	40	10.28	36	Remington	20	5.57	56	Stevens	30	6.72
17	Stevens	30	6.51	37	Remington	30	8.62	57	Remington	10	3
18	Stevens	30	9.84	38	Remington	30	8.41	58	Remington	20	5.42
19	Remington	30	6.8	39	Stevens	50	5.18	59	Remington	40	9.35
20	Stevens	10	3.85	40	Stevens	30	9.42	60	Stevens	50	11.8

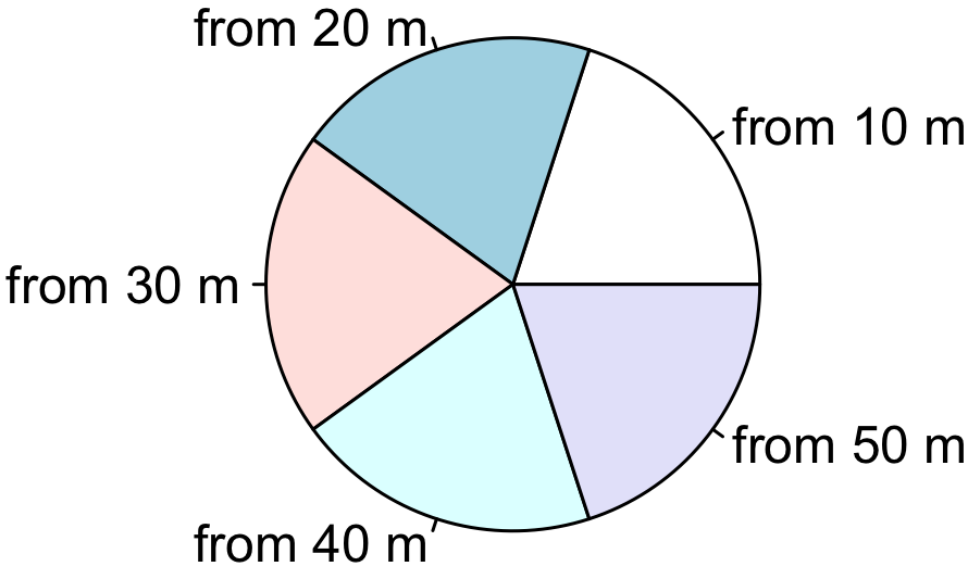
Univariate Categorical Data

- A bar chart or bar plot is a popular method for showing counts of categorical variables
  - Can be drawn horizontally or vertically
  - In a vertical bar plot (like the one shown here), height of bars indicate count and widths of bars have no meaning
  - Can potentially show multiple categorical variables on a single “stacked” bar plot



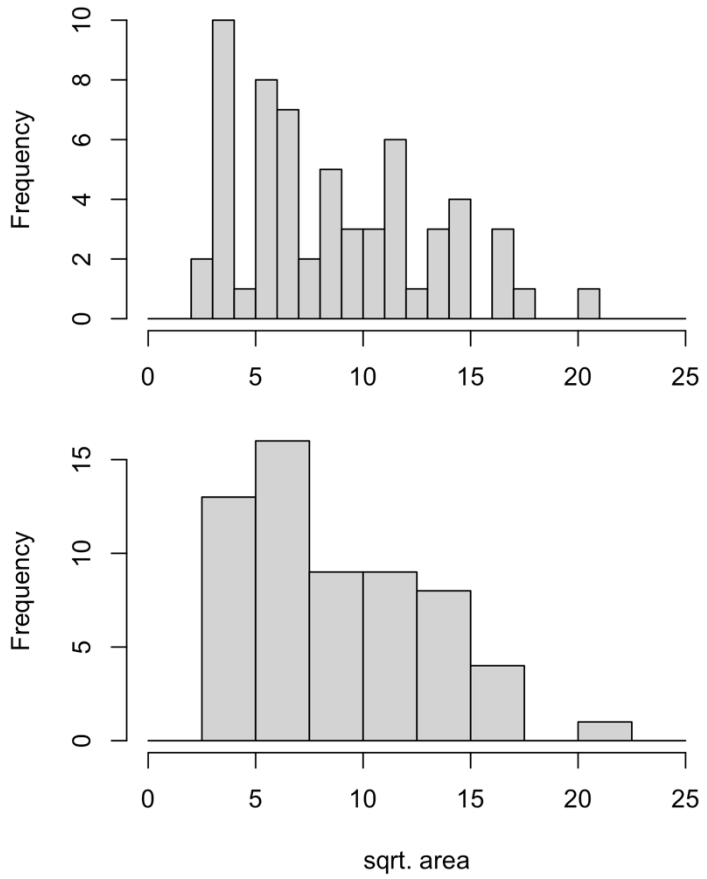
Univariate Categorical Data

- A pie chart is a popular method for showing counts of categorical variables
  - Area of pie is proportional to count
  - Can only show a single variable
- Same information can often be transmitted more precisely and as easily using a table

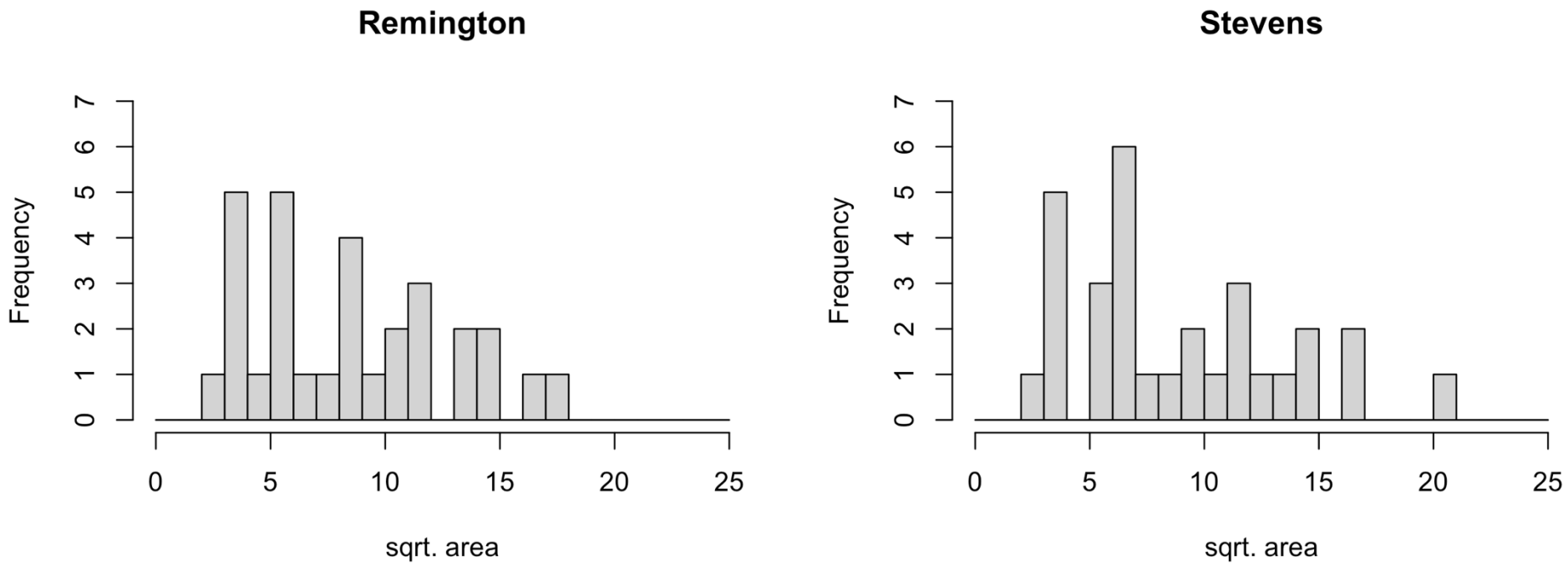


Univariate Numeric Data

- A histogram is a very popular method for summarizing the count of numeric data for a single variable
  - Most commonly vertical (as shown to the right) but can be drawn horizontally
  - Bar widths tell us something about how the numeric variable (the x-axis) is discretized, and the bar height tells us the count



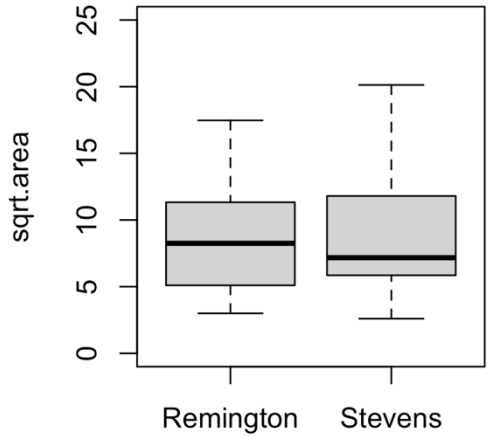
Two-dimensions (number vs category)

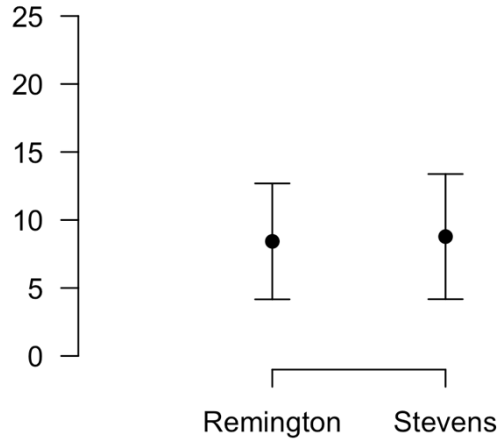


Can parse your data on categories and create several “univariate numeric data” problems.

Two-dimensions (number vs category)

- A boxplot is a good method for comparing numeric data with respect to a categories
- Sometimes referred to as a box and whisker plot
- Each “box and whisker” is constructed using summary statistics (based on the histogram) for the numeric variable for each category





Two-dimensions (category vs category)

- A contingency table is a good method for comparing the counts of two categorical variables in the data set

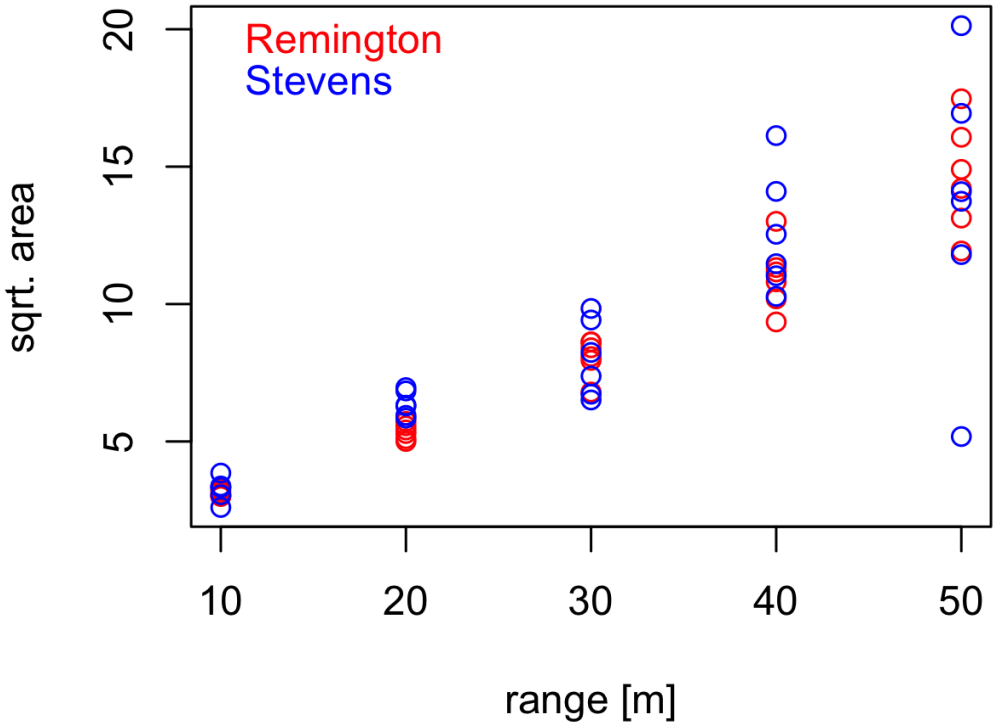
```
> table(population$range,population$gun)

    Remington Stevens
  10         6       6
  20         6       6
  30         6       6
  40         6       6
  50         6       6
>
```

Two-dimensions (number vs number)

- A scatter plot is a good method for comparing two numeric properties of the observations in the data set

- Can also display more information if you are creative with marker shapes, sizes, and colours
  - Numeric Dimension 1 = x position
  - Numeric Dimension 2 = y position
  - Numeric Dimension 3 = marker size
  - Categorical Dimension 1 = marker colour
  - Categorical Dimension 2 = marker shape

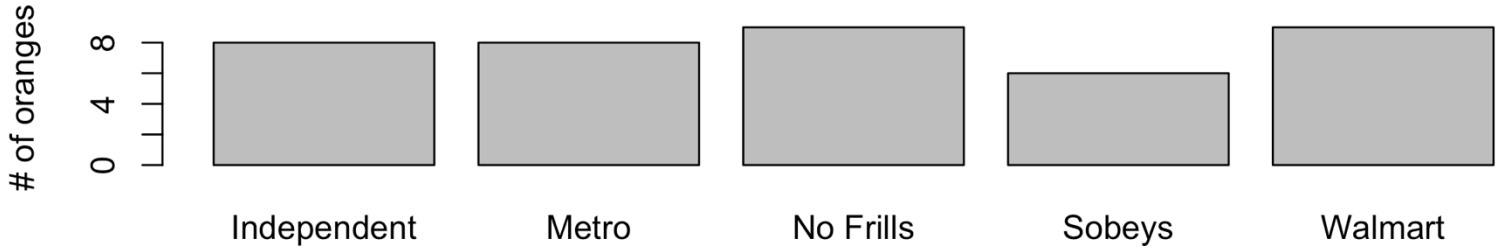


Case Study 2: Which store in Peterborough sells the best oranges?

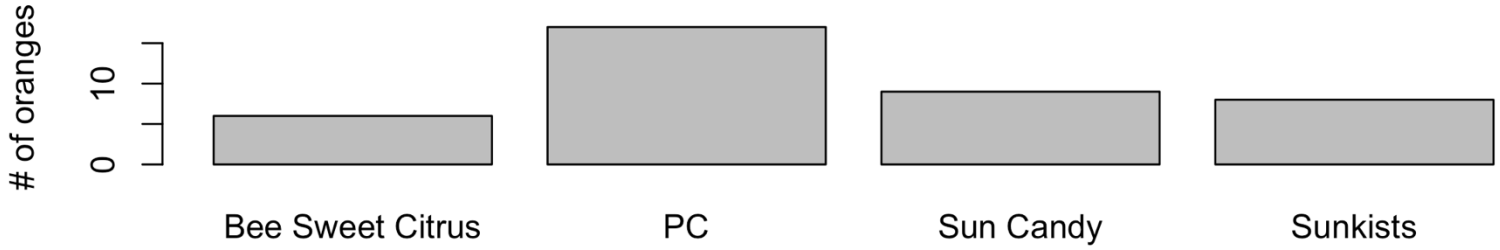
```
> head(processed_data)
  Observation Brand       Store Price.Of.Orange Mass.1 Mass.2 Max.Diameter.1 Max.Diameter.2 Min.Diameter.1
1          A1    PC Independent            0.75    191    190          73.48          72.99          68.88
2          A2    PC Independent            0.75    207    207          73.25          74.47          71.53
3          A3    PC Independent            0.75    188    188          70.99          71.24          68.45
4          A4    PC Independent            0.75    194    194          72.12          72.49          70.91
5          A5    PC Independent            0.75    186    186          72.65          71.33          66.02
6          A6    PC Independent            0.75    178    178          70.39          70.89          69.14
  Min.Diameter.2         Colour Imperfections avg.mass approx.vol est.density price_per_100g
1          69.79 Vitamin Orange            21    190.5   188.8307    1.008840      0.3937008
2          71.11   Acorn Squash            12    207.0   198.6252    1.042164      0.3623188
3          69.34   Acorn Squash            33    188.0   177.4479    1.059466      0.3989362
4          71.85        Citrusy             9    194.0   191.5094    1.013005      0.3865979
5          66.54        Citrusy            13    186.0   174.5244    1.065753      0.4032258
6          68.74 Vitamin Orange             7    178.0   177.7917    1.001172      0.4213483
```

Univariate Categorical Data

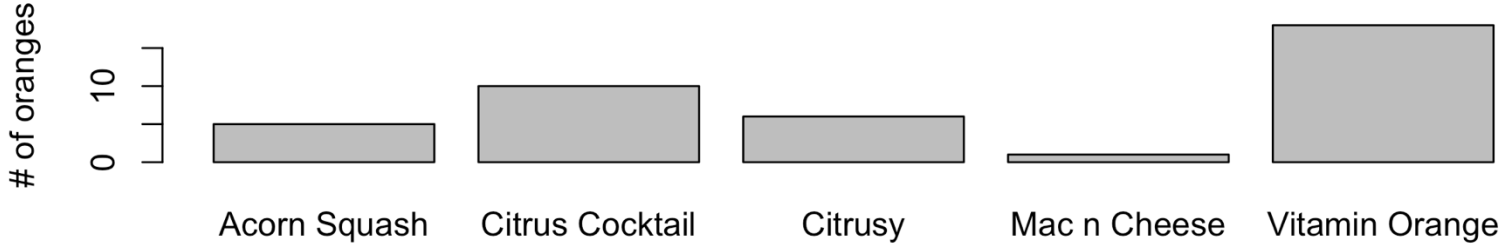
By Store



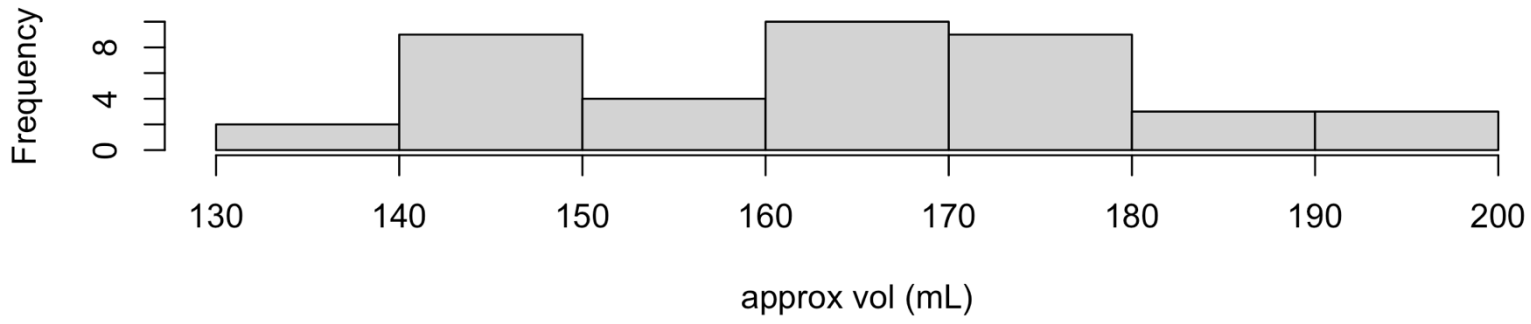
By Brand

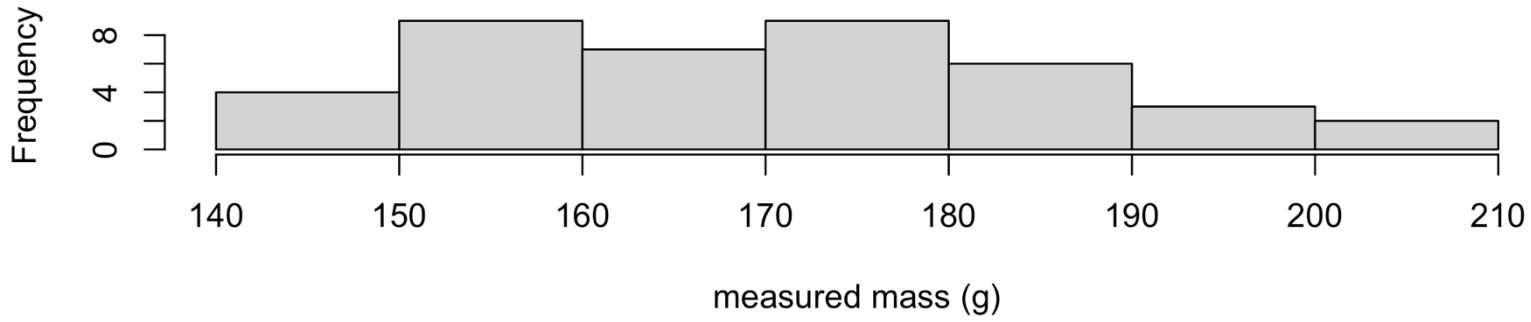


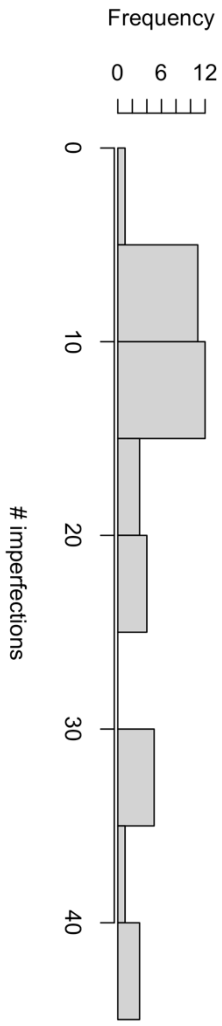
By Colour



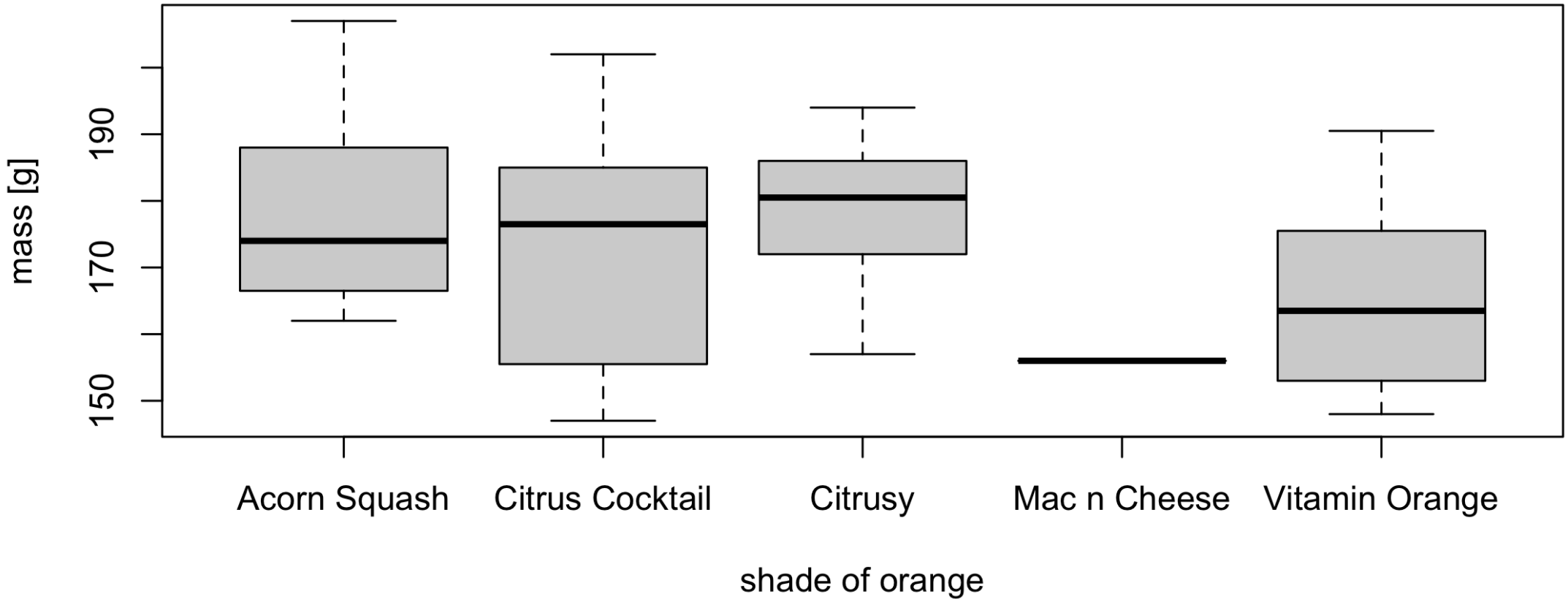
Univariate Numerical Data



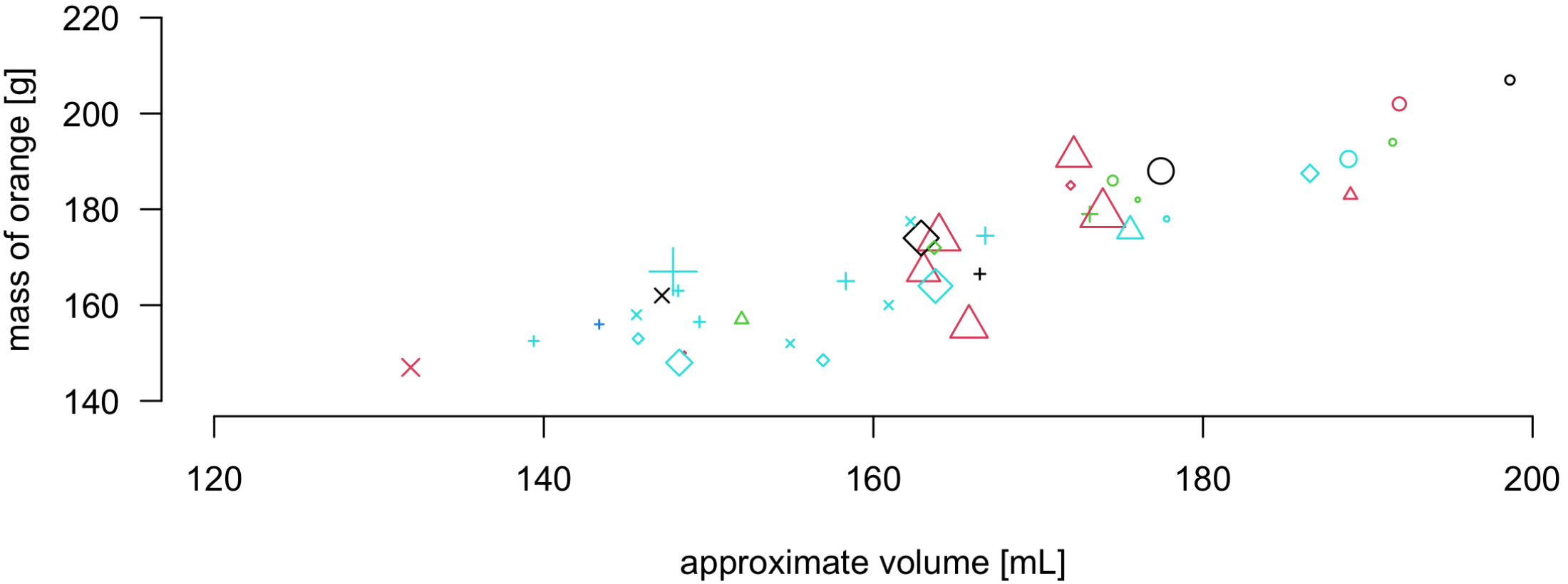




Two-dimensions (number vs. category)



Multivariate mixed data



A scatter plot of 40 oranges studied by the 2025 MScFS students in FRSC 5003: position on the x-axis is the orange's approximate volume, position on the y-axis is its measured mass, the shape of the symbol indicates from what store it was purchased, the symbol color indicates its shade of orange, and the symbol size is proportional to the observed number of imperfections.

Why visualize?

- “A good graph is almost better than any statistical analysis” – J.M. Curran
- “Graphics reveal data. Indeed, graphics can be more precise and revealing than conventional statistical computation” – E.R. Tufte
- Graphics (drawings, plots, etc.) all help us analyze data AND communicate results
- **Important:** A good graphic, like a good statistical analysis, requires good data

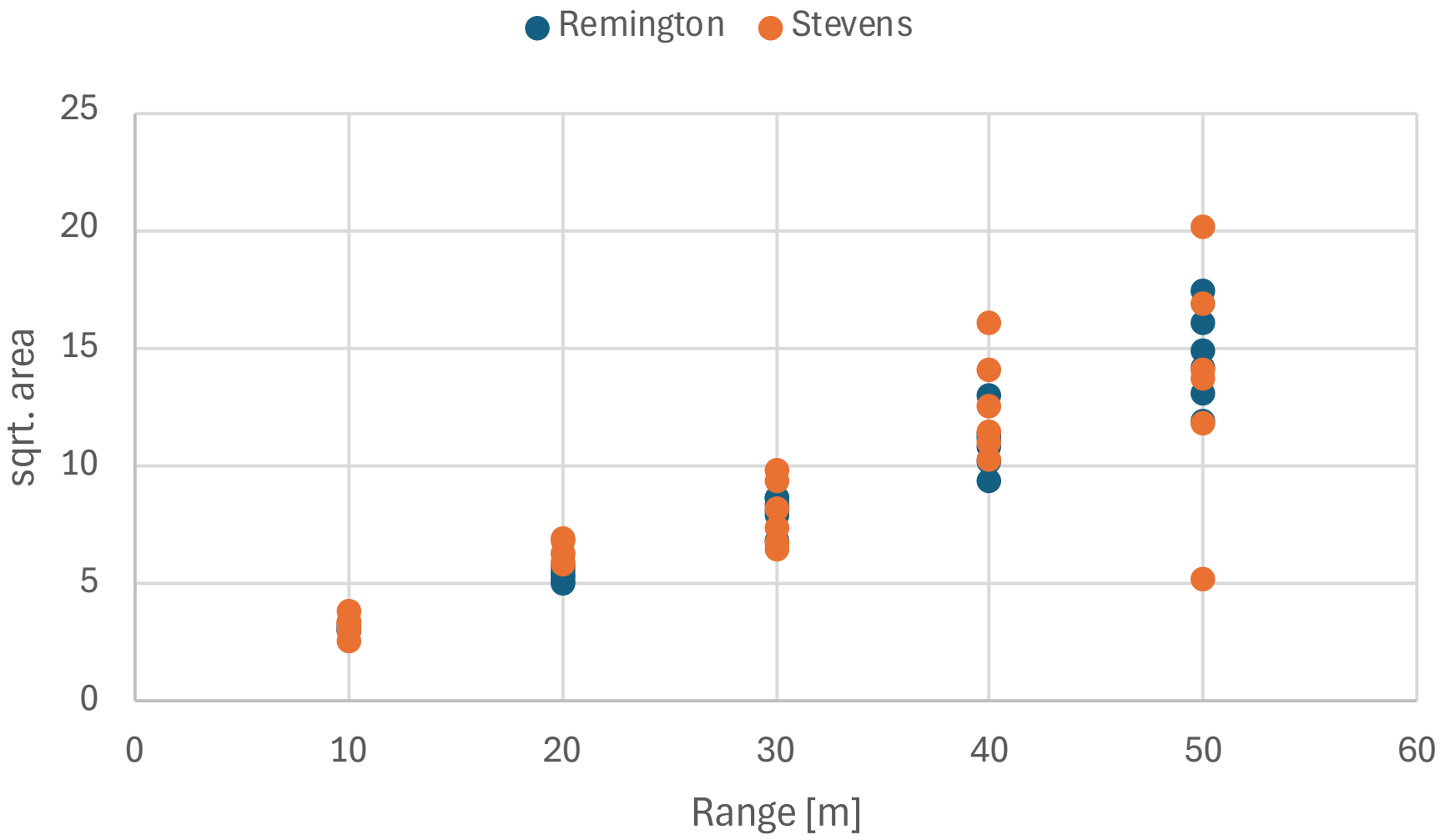
Tools for data visualization

- All spreadsheet software (e.g., MS Excel, Google Sheets, Numbers) will have some basic plotting functionality
  - Be careful that the software is plotting what you think its plotting (recommended plots can be misleading)
  - Read the documentation carefully

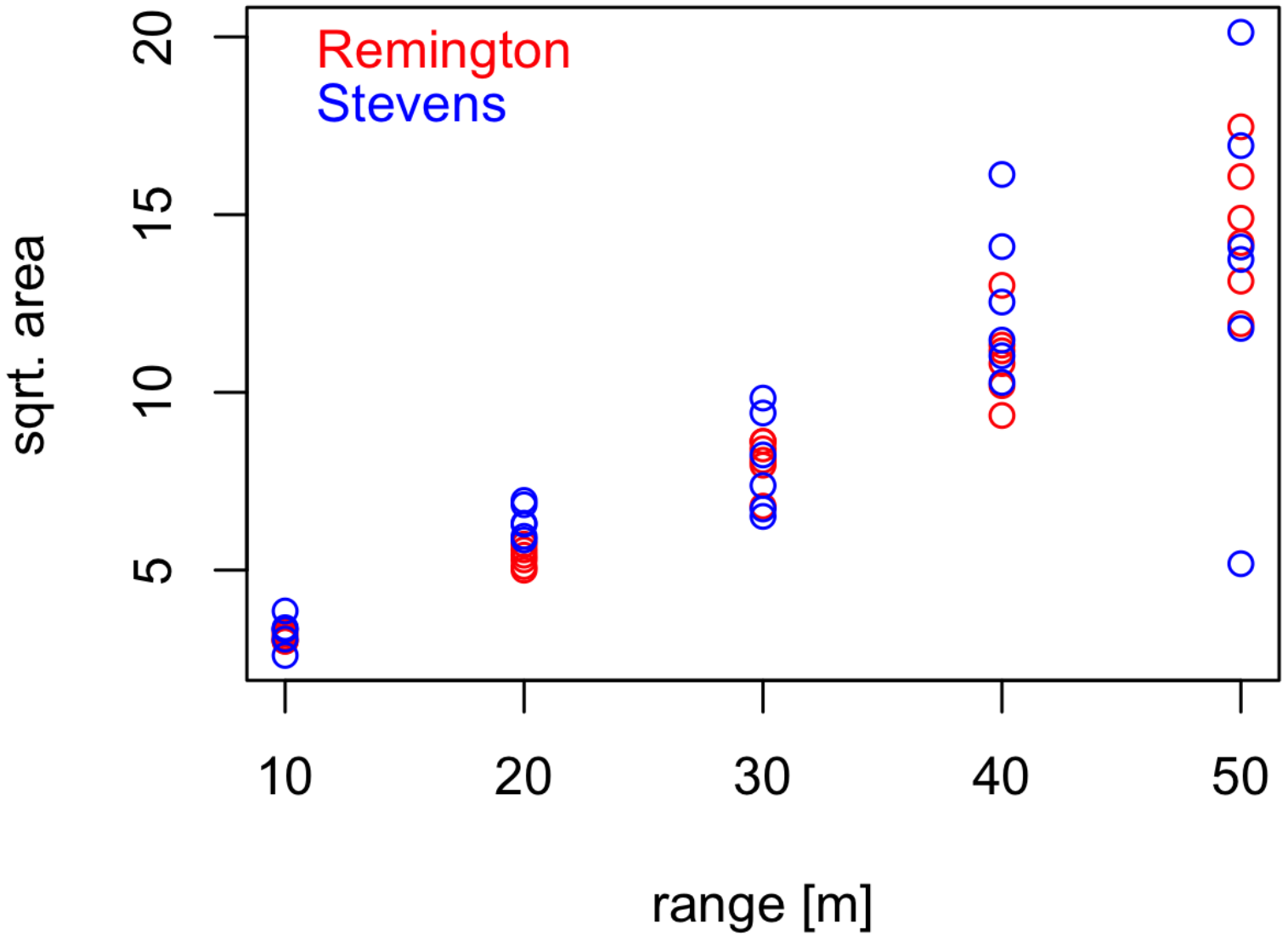
- Most programming languages (e.g., R, Python, Matlab) will have one or more “plotting” or “graphics” library so you can create plots with code

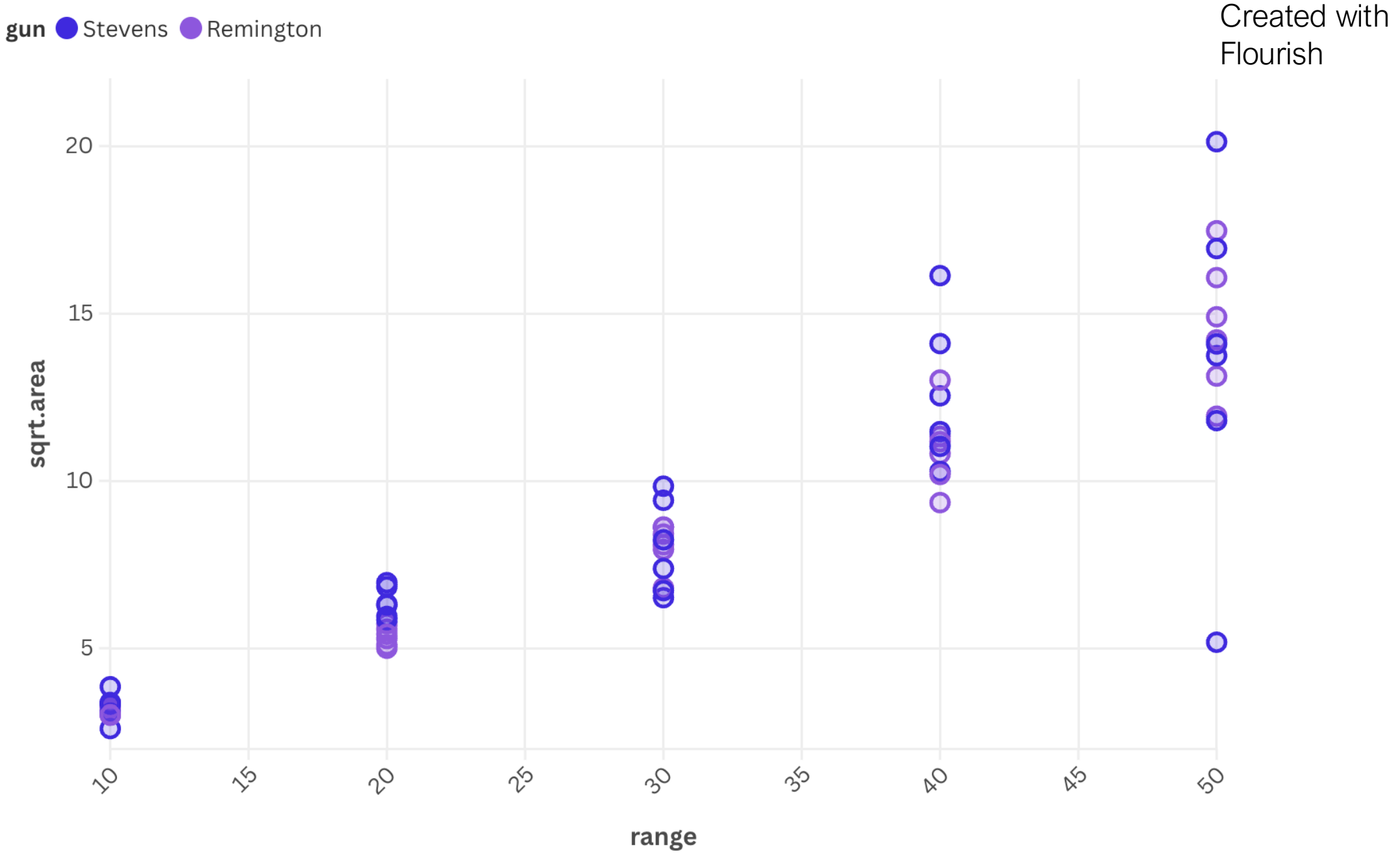
- Several free cloud-based AI tools that allow you to create visuals
  - Be careful to verify whether your data is secure and private

Created with
Microsoft Excel



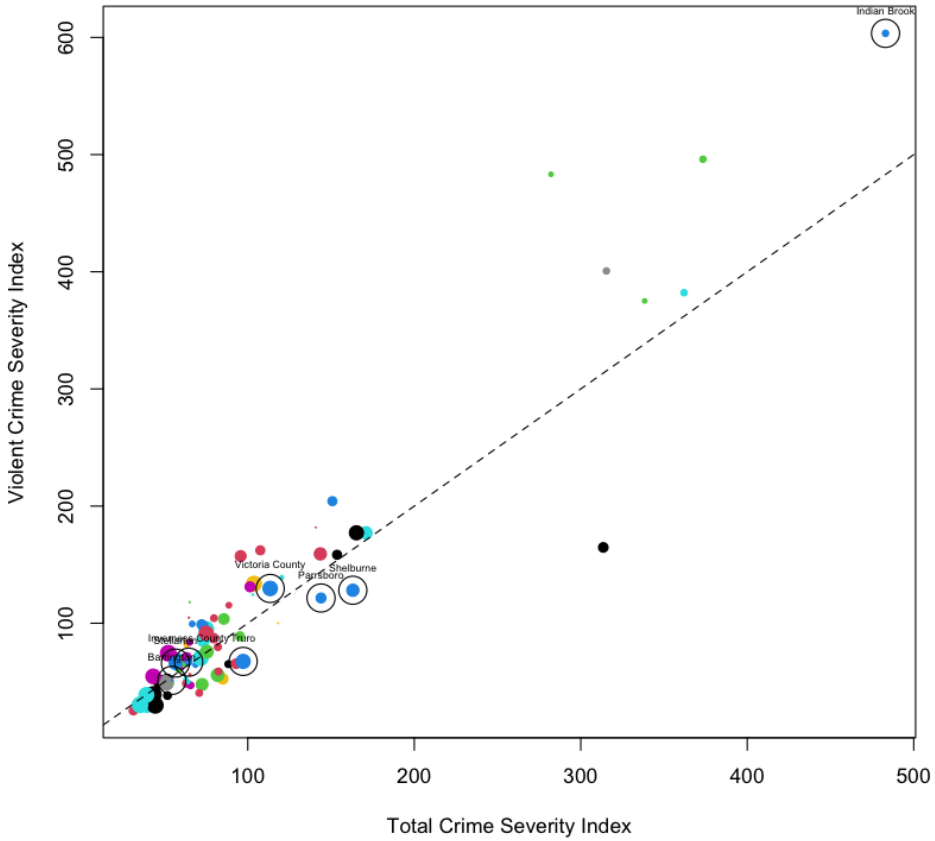
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Enhancing plots

- Several ways we can make plots **more informative**:
  - Legends
  - Lines of fit
    - Linear
    - Curves
    - Loess
  - Colors
  - Texts, arrows, highlights
- Figure captions and text descriptions are still essential



A review of probability

- **Probability:** a formal measure of the certainty that a particular event or outcome will occur.
  - **Theoretical Probability:** calculated through analysis of potential outcomes in a specific situation. Sometimes referred to as **aleatoric probability**.
    - E.g., Probability of flipping a head with a *fair coin* is 1/2. The probability of rolling the same number with a pair of *fair dice* is 1/6.
  - **Empirical Probability:** estimated from past measurements or observations (using statistical tools).
    - E.g., As per available historical data, the probability of an eye-witness making an incorrect observation is 0.35. As per available historical data, the probability of it snowing in August in Peterborough is 0.0001. (I made these two numbers up...)

Calculating probabilities

- The fundamental assumption when calculating theoretical probabilities is that we know all possible outcomes for the given situation
  - Examples: There are 2 possible outcomes when flipping a coin. There are 6 possible outcomes when rolling a die.

$$\text{Probability} = \frac{\text{number of selected outcomes}}{\text{total number of possible outcomes}}$$

- Computed probability can be reported as a fraction, decimal or percentage, as appropriate

Calculating probabilities

Example 1: A deck of playing cards contains four suits – spades and clubs, which are black, and diamonds and hearts, which are red. Each suit contains an ace, nine numbered cards (2-10) and three royal cards – king, queen and jack. There are 52 cards in a deck. The cards are fully shuffled to randomize their order before each experiment. Calculate the probability of drawing:

a) A red card
b) A club
c) A royal card

Calculating probabilities

Solution 1a: There are 26 red cards in a deck (13 hearts and 13 diamonds). The probability of drawing a red card is therefore:

$$P(\text{red}) = \frac{26}{52} = \frac{1}{2} = 0.5$$

Calculating probabilities

Solution 1b: There are 13 clubs in a deck. The probability of drawing a club is therefore:

$$P(\text{club}) = \frac{13}{52} = \frac{1}{4} = 0.25$$

Calculating probabilities

Solution 1c: There are 3 royal cards (Jack, Queen, King) per suit (club, spade, diamond, heart) in a deck. The probability of drawing a royal is therefore:

$$P(\text{royal}) = \frac{3 \times 4}{52} = \frac{3}{13} = 0.23$$

Calculating odds

- **Odds:** another way to discuss probabilities, often used in betting as well as forensic science

- Calculated as the ratio between the probability of an event A occurring and its complement (the probability of the event not happening)

$$odds = \frac{P(\mathrm{A})}{P(\overline{\mathrm{A}})}$$

- Example: The odds of being dealt a royal card are $\frac{0.23}{1-0.23} = 0.3$

Combining probabilities

- There are some useful shortcuts for calculating probabilities of multiple *independent events* occurring (or not occurring)

Rule 1. The probability of two independent events **A and B happening** is given by: $P(\mathrm{A\ and\ B}) = P(\mathrm{A}) \times P(\mathrm{B})$.

Rule 2. The probability of two independent events **A or B happening** is given by: $P(\mathrm{A\ or\ B}) = P(\mathrm{A}) + P(\mathrm{B})$.

Rule 3. The probability of an event **A not happening** is the complement to the probability of the event happening, and can be calculated as: $P(\overline{\mathrm{A}}) = 1 - P(\mathrm{A})$.

Combining probabilities

Example 2: You have a bag with 8 green marbles and 4 red marbles (12 marbles total). You shake the bag up, pull a marble out of the bag and record its colour. You put the marble back, shake the bag up, and then pull a marble out of the bag and record its colour. What is the probability that

a) you pulled a green marble and then a red marble?
b) you pulled two marbles of the same colour?
c) you pulled one marble of each colour?

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c) you pulled one marble of each colour?

Combining probabilities

Solution 2a: From **Rule 1**, the probability of pulling a green and red marble is

$$P(\text{green and red}) = P(\text{green}) \times P(\text{red}) = \frac{2}{3} \times \frac{1}{3} = \frac{2}{9} = 0.22$$

Example 2: You have a bag with 8 green marbles and 4 red marbles (12 marbles total). You shake the bag up, pull a marble out of the bag and record its colour. You put the marble back, shake the bag up, and then pull a marble out of the bag and record its colour. What is the probability that

a) you pulled a green marble and then a red marble?
b) you pulled two marbles of the same colour?
c) you pulled one marble of each colour?

Combining probabilities

Solution 2b: From **Rule 1**, the probability of pulling two green marbles is

$$P(\text{G and G}) = P(\text{G}) \times P(\text{G}) = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9} = 0.44$$

And the probability of pulling two red marbles is

$$P(\text{R and R}) = P(\text{R}) \times P(\text{R}) = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9} = 0.11$$

From **Rule 2**, the probability of pulling two marbles of the same color is:

$$P\big((\text{G and G}) \text{ OR } (\text{R and R})\big) = P(\text{G and G}) + P(\text{R and R}) = \frac{4}{9} + \frac{1}{9} = \frac{5}{9} = 0.55$$

Example 2: You have a bag with 8 green marbles and 4 red marbles (12 marbles total). You shake the bag up, pull a marble out of the bag and record its colour. You put the marble back, shake the bag up, and then pull a marble out of the bag and record its colour. What is the probability that

a) you pulled a green marble and then a red marble?
b) you pulled two marbles of the same colour?
c) you pulled one marble of each colour?

Combining probabilities

Solution 2c: Can repeat process from previous example to end up with

$$P\big((\text{G and R})\ \text{OR}\ (\text{R and G})\big) = P(\text{G and R}) + P(\text{R and G}) = \frac{2}{9} + \frac{2}{9} = \frac{4}{9} = 0.44$$

Alternatively, you can observe that probability of pulling one marble of each colour is the **complement** to the probability of pulling two marbles of the same colour, and leverage **Rule 3**:

$$P\big((\text{G and R})\ \text{OR}\ (\text{R and G})\big) = 1 - P\big((\text{G and G})\ \text{OR}\ (\text{R and R})\big) = 1 - \frac{5}{9} = \frac{4}{9}0.44$$

Combining probabilities

Example 3: The distribution of men's shoe sizes in a large hypothetical city is shown below.

Size	5	6	7	8	9	10	11	12
%	7	11	10	14	16	18	16	8

Calculate the probability that

a) a man selected at random will have a shoe size of 9 or 10.

b) two men selected at random will have the same size shoe.

c) two men selected at random will not have the same size shoe.

Example 3: The distribution of men's shoe sizes in a large hypothetical city is shown below.

Size	5	6	7	8	9	10	11	12
%	7	11	10	14	16	18	16	8

Calculate the probability that

a) a man selected at random will have a shoe size of 9 or 10.

b) two men selected at random will have the same size shoe.

c) two men selected at random will not have the same size shoe.

Combining probabilities

Solution 3a: From **Rule 2**, the probability of a man from this hypothetical city having a shoe size of 9 or 10 is

$$P(9 \text{ or } 10) = P(9) + P(10) = \frac{16}{100} + \frac{18}{100} = 0.34$$

Example 3: The distribution of men's shoe sizes in a large hypothetical city is shown below.

Size	5	6	7	8	9	10	11	12
%	7	11	10	14	16	18	16	8

Calculate the probability that

a) a man selected at random will have a shoe size of 9 or 10.

b) two men selected at random will have the same size shoe.

c) two men selected at random will not have the same size shoe.

Combining probabilities

Solution 3b: We have 8 cases that will result in the two men having the same shoe size: if both men wear size 5, size 6, size 7, and so on to size 12. Using **Rules 1 and 2**, we can calculate:

$$P(5 \text{ and } 5) = P(5) \times P(5) \qquad P(9 \text{ and } 9) = P(9) \times P(9)$$
$$P(6 \text{ and } 6) = P(6) \times P(6) \qquad P(10 \text{ and } 10) = P(10) \times P(10)$$
$$P(7 \text{ and } 7) = P(7) \times P(7) \qquad P(11 \text{ and } 11) = P(11) \times P(11)$$
$$P(8 \text{ and } 8) = P(8) \times P(8) \qquad P(12 \text{ and } 12) = P(12) \times P(12)$$

$$P(\text{same size}) = P\big((5 \text{ and } 5) \text{ or } \ldots \text{ or } (12 \text{ and } 12)\big) = P(5 \text{ and } 5) + \cdots + P(12 \text{ and } 12)$$

Combining probabilities

Example 3: The distribution of men's shoe sizes in a large hypothetical city is shown below.

Size	5	6	7	8	9	10	11	12
%	7	11	10	14	16	18	16	8

Calculate the probability that

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c) two men selected at random will not have the same size shoe.

Solution 3b: We have 8 cases that will result in the two men having the same shoe size: if both men wear size 5, size 6, size 7, and so on to size 12. Using **Rules 1 and 2**, we can calculate:

$$P(5 \text{ and } 5) = 4.9\times10^{-3} \qquad P(9 \text{ and } 9) = 2.56\times10^{-2}$$

$$P(6 \text{ and } 6) = 1.21\times10^{-2} \qquad P(10 \text{ and } 10) = 3.24\times10^{-2}$$

$$P(7 \text{ and } 7) = 1\times10^{-2} \qquad P(11 \text{ and } 11) = 2.56\times10^{-2}$$

$$P(8 \text{ and } 8) = 1.96\times10^{-2} \qquad P(12 \text{ and } 12) = 6.4\times10^{-3}$$

$$P(\text{same size}) = 0.13$$

Combining probabilities

Example 3: The distribution of men's shoe sizes in a large hypothetical city is shown below.

Size	5	6	7	8	9	10	11	12
%	7	11	10	14	16	18	16	8

Calculate the probability that

a) a man selected at random will have a shoe size of 9 or 10.

b) two men selected at random will have the same size shoe.

c) two men selected at random will not have the same size shoe.

Solution 3c: From **Rule 3**, the probability of two men from this hypothetical city selected at random not having the same shoe size is

$$P(\text{not same size}) = 1 - P(\text{same size}) = 0.87$$

Conditional probabilities

- When two events are not independent, we can calculate conditional probabilities

- Notation for conditional probability of A given B: $P(\mathrm{A}|\mathrm{B})$

- NOTE: $P(\mathrm{A}|\mathrm{B})$ is not necessarily the same as $P(\mathrm{B}|\mathrm{A})$

Conditional probabilities

Example 4: You have a bag with 8 green marbles and 4 red marbles (12 marbles total). You shake the bag up, pull a marble out of the bag and record its colour. You shake the bag up again, and then pull another marble out of the bag and record its colour. What is the probability that

a) The first marble is green and the second marble is red
b) Both marbles are green
c) **The first marble is red and the second marble is red or green**

Conditional probabilities

Example 4: You have a bag with 8 green marbles and 4 red marbles (12 marbles total). You shake the bag up, pull a marble out of the bag and record its colour. You shake the bag up again, and then pull another marble out of the bag and record its colour. What is the probability that

a) The first marble is green and the second marble is red
b) Both marbles are green
c) The first marble is red and the second marble is red or green

Solution 4a: The probability that the first marble is green is:

$$P(\text{first} = \text{green}) = \frac{8}{12} = \frac{2}{3}$$

The probability that the second marble is red, given that the first one was green is:

$$P(\text{second} = \text{red} \mid \text{first} = \text{green}) = \frac{4}{11}$$

Therefore:

$$P(\text{first} = \text{green AND second} = \text{red}) = \frac{2}{3} \times \frac{4}{11} = 0.242$$

Conditional probabilities

Example 4: You have a bag with 8 green marbles and 4 red marbles (12 marbles total). You shake the bag up, pull a marble out of the bag and record its colour. You shake the bag up again, and then pull another marble out of the bag and record its colour. What is the probability that

a) The first marble is green and the second marble is red
b) Both marbles are green
c) The first marble is red and the second marble is red or green

Solution 4b: The probability that the first marble is green is:

$$P(\text{first} = \text{green}) = \frac{8}{12} = \frac{2}{3}$$

The probability that the second marble is green, given that the first one was green is:

$$P(\text{second} = \text{green} \mid \text{first} = \text{green}) = \frac{7}{11}$$

Therefore:

$$P(\text{first} = \text{green AND second} = \text{green}) = \frac{2}{3} \times \frac{7}{11} = 0.424$$

Conditional probabilities

Example 4: You have a bag with 8 green marbles and 4 red marbles (12 marbles total). You shake the bag up, pull a marble out of the bag and record its colour. You shake the bag up again, and then pull another marble out of the bag and record its colour. What is the probability that

a) The first marble is green and the second marble is red
b) Both marbles are green
c) The first marble is red and the second marble is red or green

Solution 4c: The probability that the first marble is red is:

$$P(\text{first} = \text{red}) = \frac{4}{12} = \frac{1}{3}$$

The probability that the second marble is green, given that the first one was red is:

$$P(\text{second} = \text{green} \mid \text{first} = \text{red}) = \frac{8}{11}$$

The probability that the second marble is green, given that the first one was red is:

$$P(\text{second} = \text{red} \mid \text{first} = \text{red}) = \frac{3}{11}$$

Example 4: You have a bag with 8 green marbles and 4 red marbles (12 marbles total). You shake the bag up, pull a marble out of the bag and record its colour. You shake the bag up again, and then pull another marble out of the bag and record its colour. What is the probability that

a) The first marble is green and the second marble is red
b) Both marbles are green
c) The first marble is red and the second marble is red or green

Conditional probabilities

Solution 4c: Therefore:

$$P(\text{first} = \text{red AND (second} = \text{green or second} = \text{red})) = \frac{1}{3} \times \left(\frac{8}{11} + \frac{3}{11} \right) = 0.33$$

Note: P(GR) = 0.242, P(GG) = 0.424, therefore

$$P(\text{RG or RR}) = 1 - 0.242 - 0.424 = 0.33$$

Probability distributions

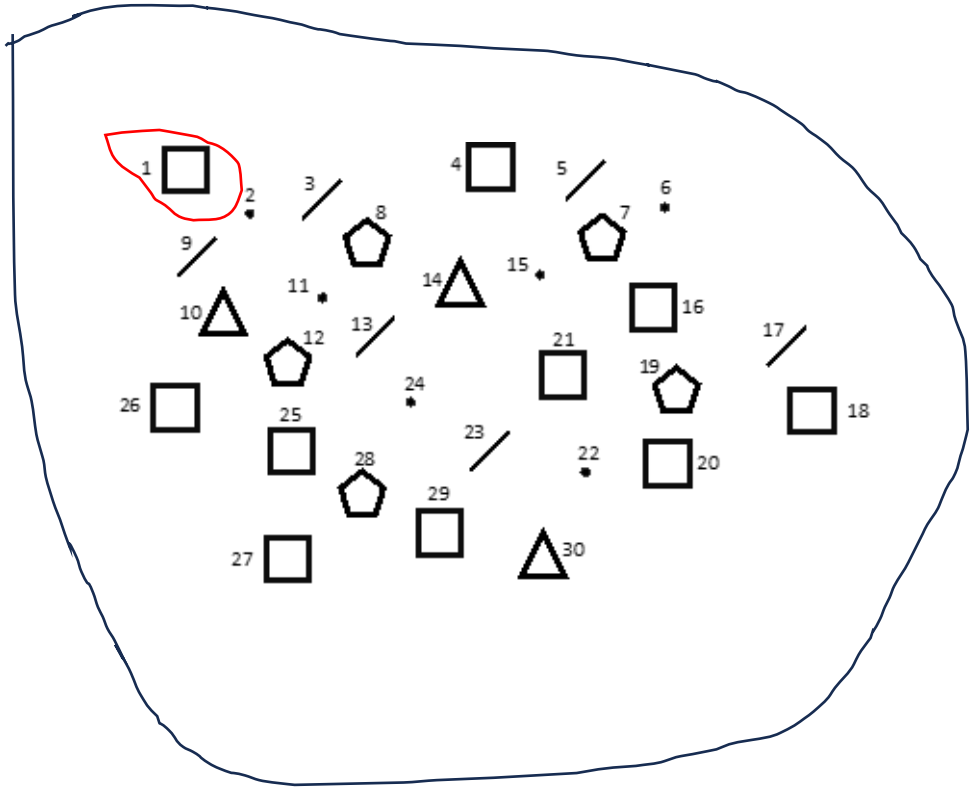
- **Probability distribution:** A mathematical function that maps between specific outcomes/events (inputs) and the probability of their occurrence (outputs)
  - Probability mass function: discrete inputs
  - Probability density function: continuous inputs

- Several common *well-characterized* probability distributions that we encounter in forensic science:
  - Normal Distribution
  - Binomial Distribution

- **Can be approximated from data**

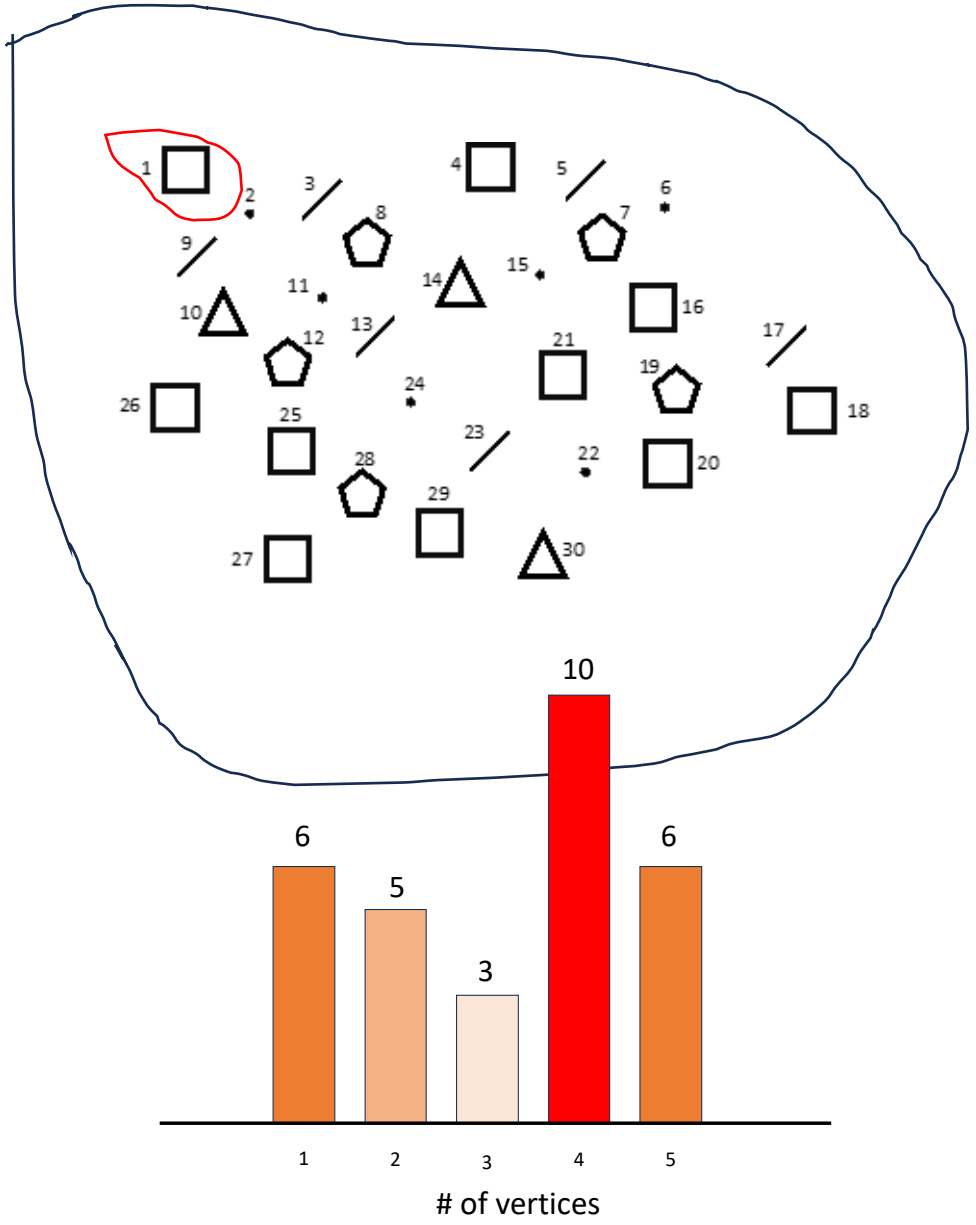
Probability and Statistics

- A **population** is a set of objects
- Every object has **properties**
- Some properties are **constant** across the population, others are **variable**



Probability and Statistics

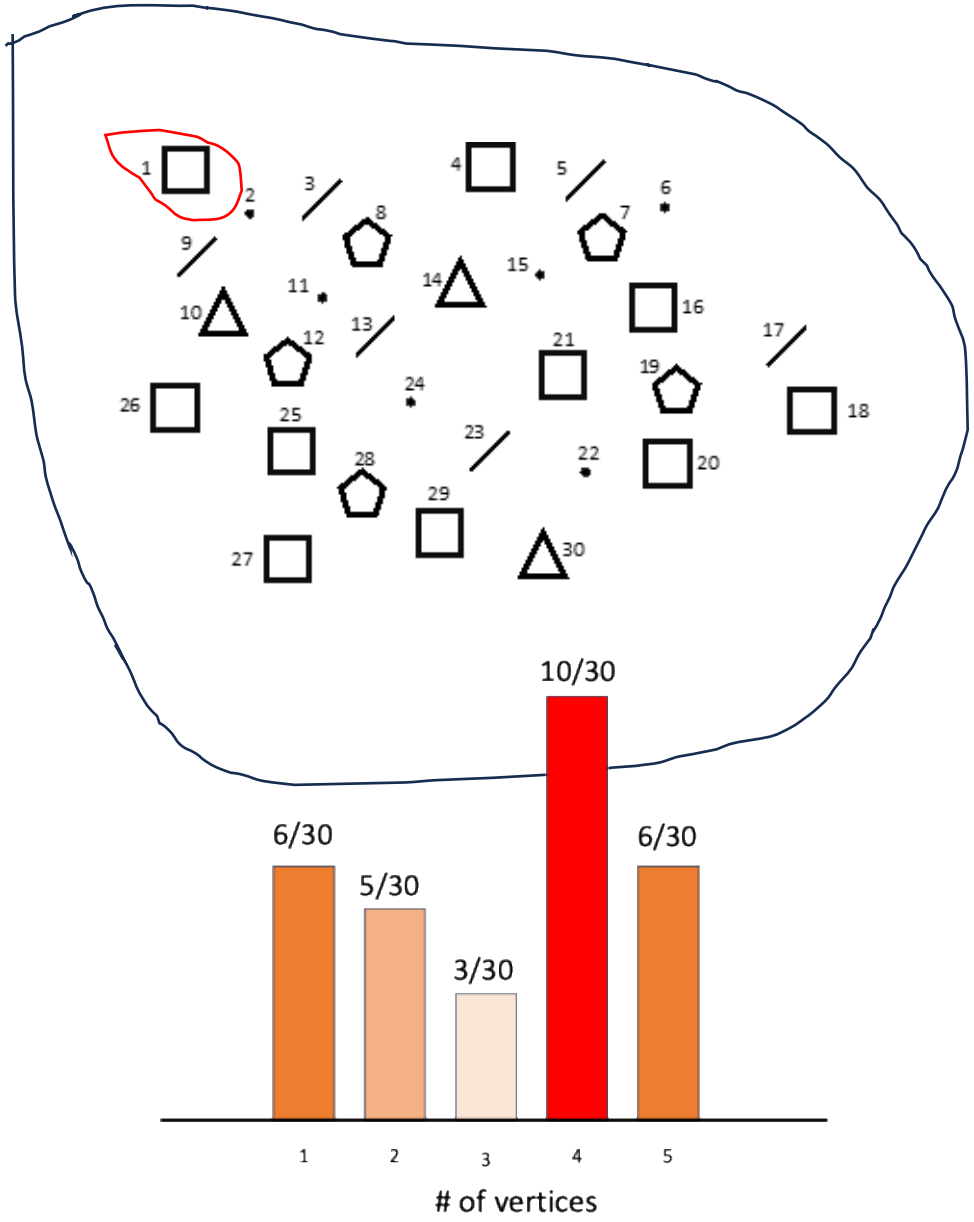
- A **histogram** is a way of visualizing a single variable from the sample (or population): Frequency Distribution



Probability and Statistics

- A **histogram** is a way of visualizing a single variable from the sample (or population): Frequency Distribution

- A histogram using relative frequencies can be interpreted as a probability distribution

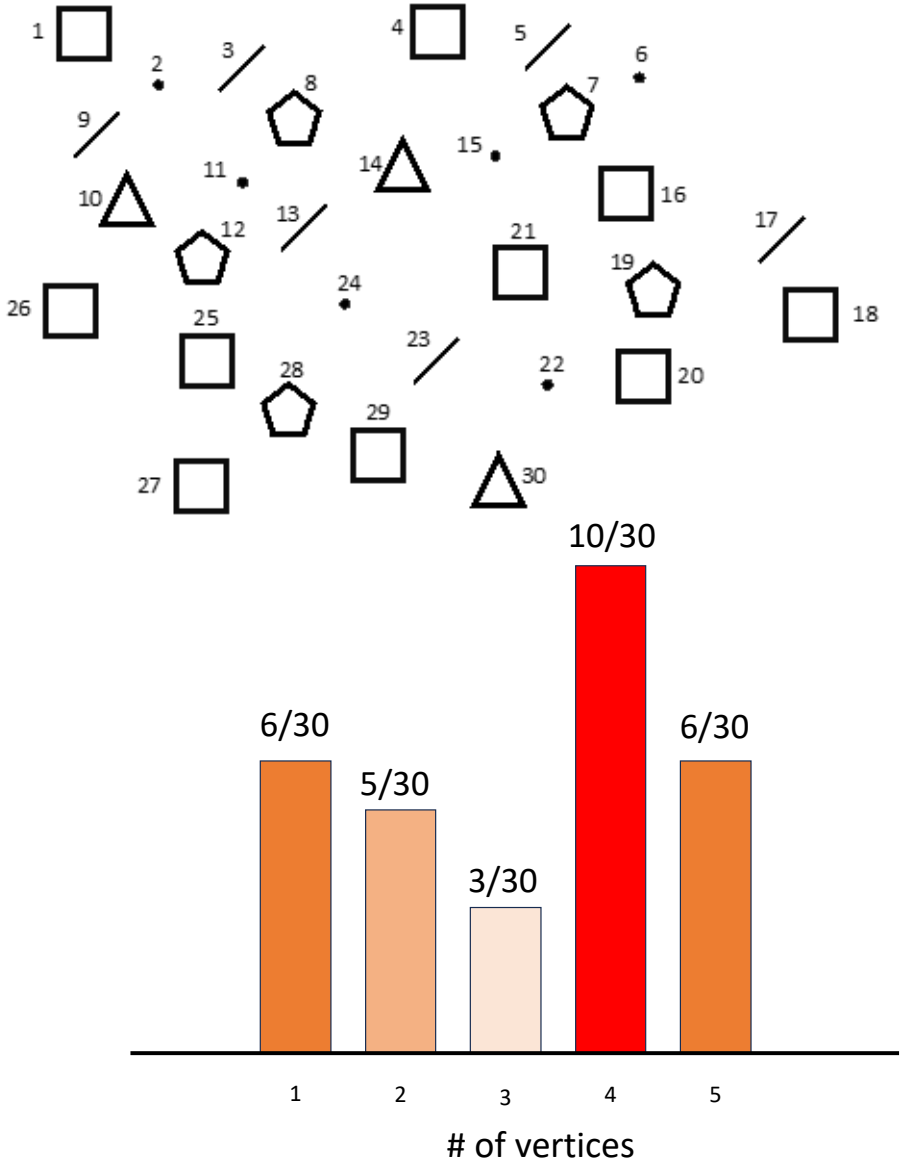


Practice Problems

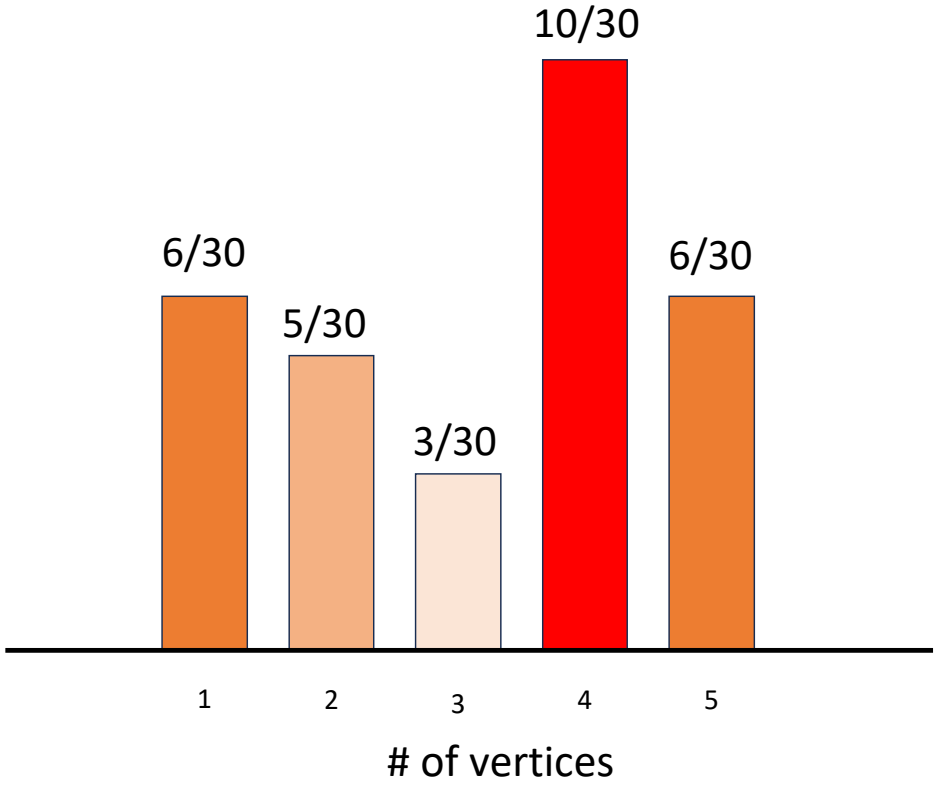
1. Selecting a shape at random, what is the probability that you'll pick a triangle?

2. Selecting a shape at random, what is the probability that you won't pick a pentagon?

3. Selecting a shape at random, what is the probability that you'll pick a triangle OR a square?



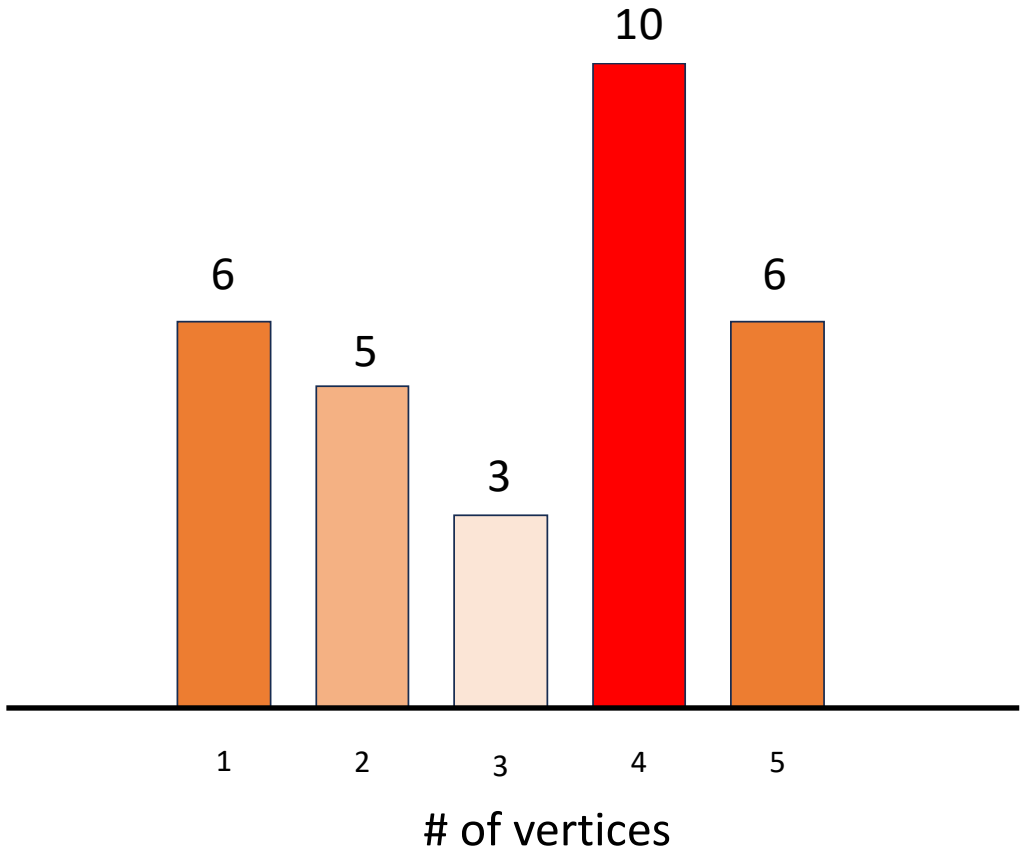
Foundations of statistics



- **Parameter:** a property/characteristic of the variable across the population; a property or characteristic of a frequency distribution of the population

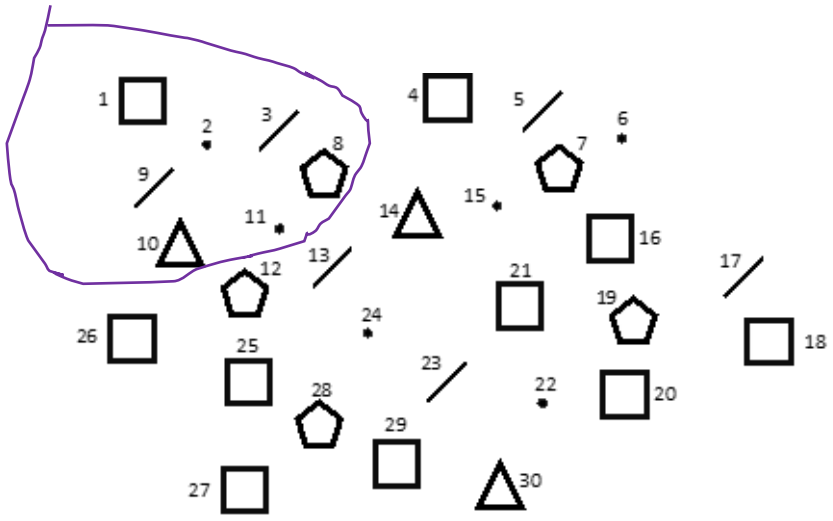
Types of parameters

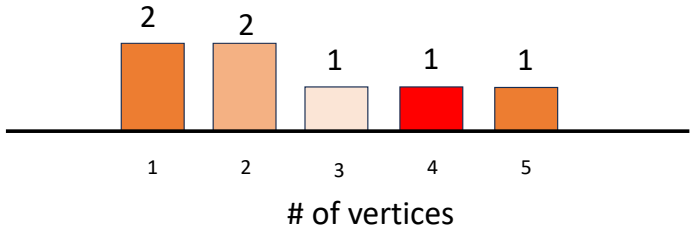
- **Central Tendency:** the "typical" value of a variable
  - E.g., mean, median, mode
- **Dispersion**: how different are values of the variable
  - E.g., standard deviation, variance, range
- **Skewness/Kurtosis:** advanced descriptions of location and spread



IMPORTANT: *Just because a parameter can be computed doesn't mean it tells us anything useful.*

Foundations of statistics





- **Statistic:** a property/characteristic of the variable across a sample; a property or characteristic of the frequency distribution of a sample

Key language to remember

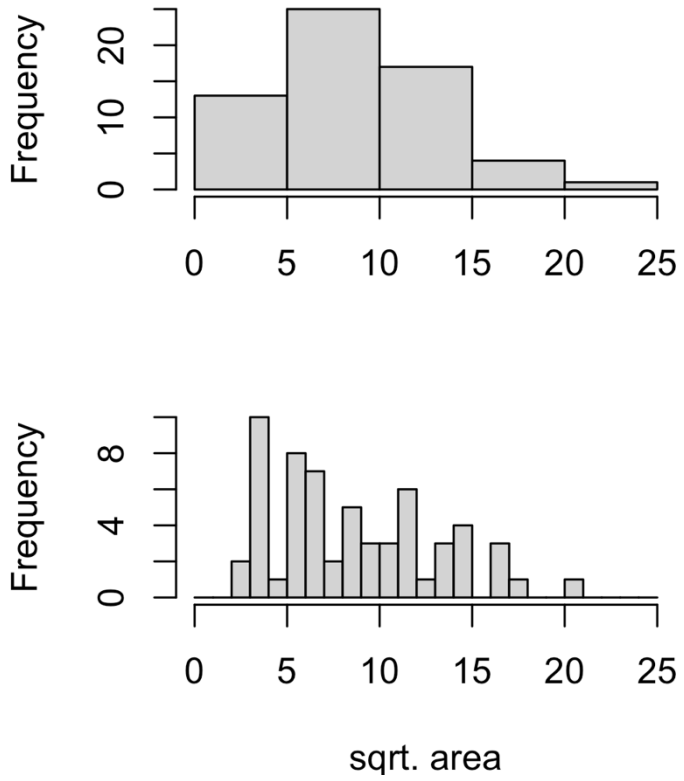
- A **property** is a characteristic by which we can describe an **object**
- A **statistic** is a characteristic by which we can describe a **sample**
- A **parameter** is a characteristic by which we can describe a **population**

Case Study 1: Does shot gun brand effect pellet scatter pattern?

Target	gun	range	sqrt.area	Target	gun	range	sqrt.area	Target	gun	range	sqrt.area
1	Stevens	40	12.54	21	Stevens	20	6.96	41	Stevens	40	11.03
2	Remington	40	13.01	22	Stevens	20	6.32	42	Stevens	30	8.24
3	Remington	50	13.13	23	Stevens	40	11.47	43	Remington	20	5.73
4	Stevens	50	13.74	24	Remington	50	17.47	44	Stevens	30	7.38
5	Remington	10	3.29	25	Stevens	20	6.84	45	Stevens	10	2.6
6	Remington	30	7.95	26	Stevens	10	3.06	46	Remington	50	16.07
7	Remington	40	10.81	27	Stevens	10	3.35	47	Remington	50	14.9
8	Stevens	40	16.13	28	Remington	40	10.19	48	Stevens	20	6.29
9	Remington	10	3.01	29	Remington	20	5	49	Stevens	50	16.94
10	Remington	20	5.1	30	Remington	30	8.09	50	Stevens	50	14.09
11	Stevens	10	3.38	31	Remington	40	11.17	51	Remington	10	3.02
12	Remington	10	3.2	32	Remington	20	5.29	52	Stevens	20	5.85
13	Remington	30	8.62	33	Remington	50	14.21	53	Stevens	20	5.95
14	Remington	10	3.11	34	Remington	50	11.93	54	Stevens	50	20.13
15	Remington	40	11.33	35	Stevens	40	14.1	55	Stevens	10	3.33
16	Stevens	40	10.28	36	Remington	20	5.57	56	Stevens	30	6.72
17	Stevens	30	6.51	37	Remington	30	8.62	57	Remington	10	3
18	Stevens	30	9.84	38	Remington	30	8.41	58	Remington	20	5.42
19	Remington	30	6.8	39	Stevens	50	5.18	59	Remington	40	9.35
20	Stevens	10	3.85	40	Stevens	30	9.42	60	Stevens	50	11.8

Computing descriptive statistics

- What is the central tendency?
  - Arithmetic Mean
  - Median
  - Mode
- What is the dispersion?
  - Range
  - Variance
  - Standard Deviation

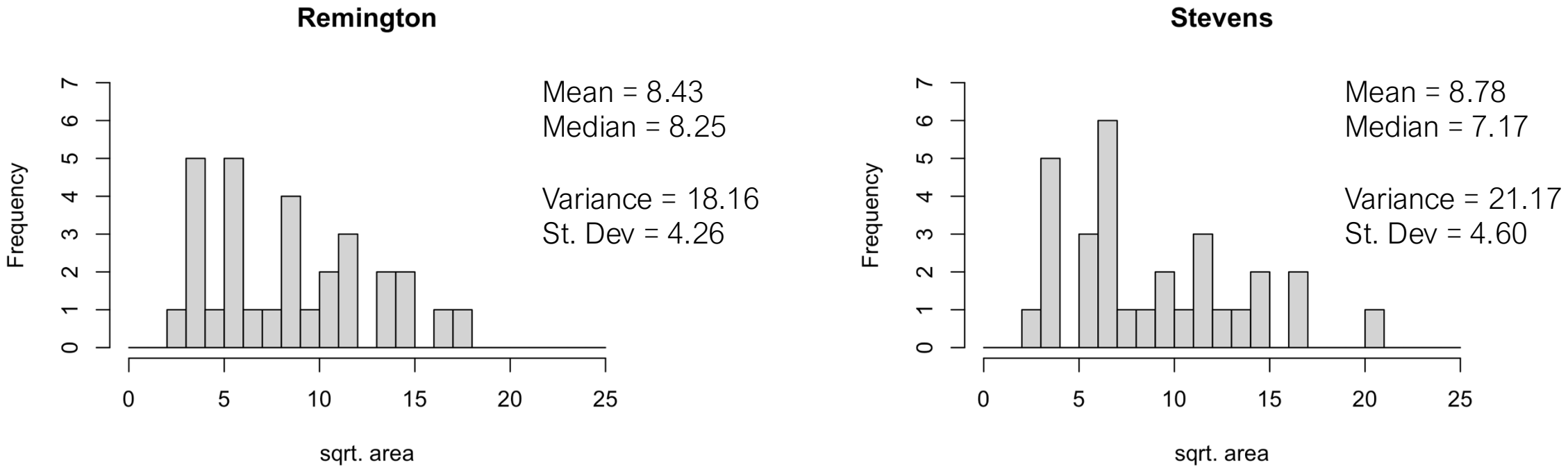


Mean = 8.602
Median = 8.02

Variance = 19.36
St. Dev = 4.40

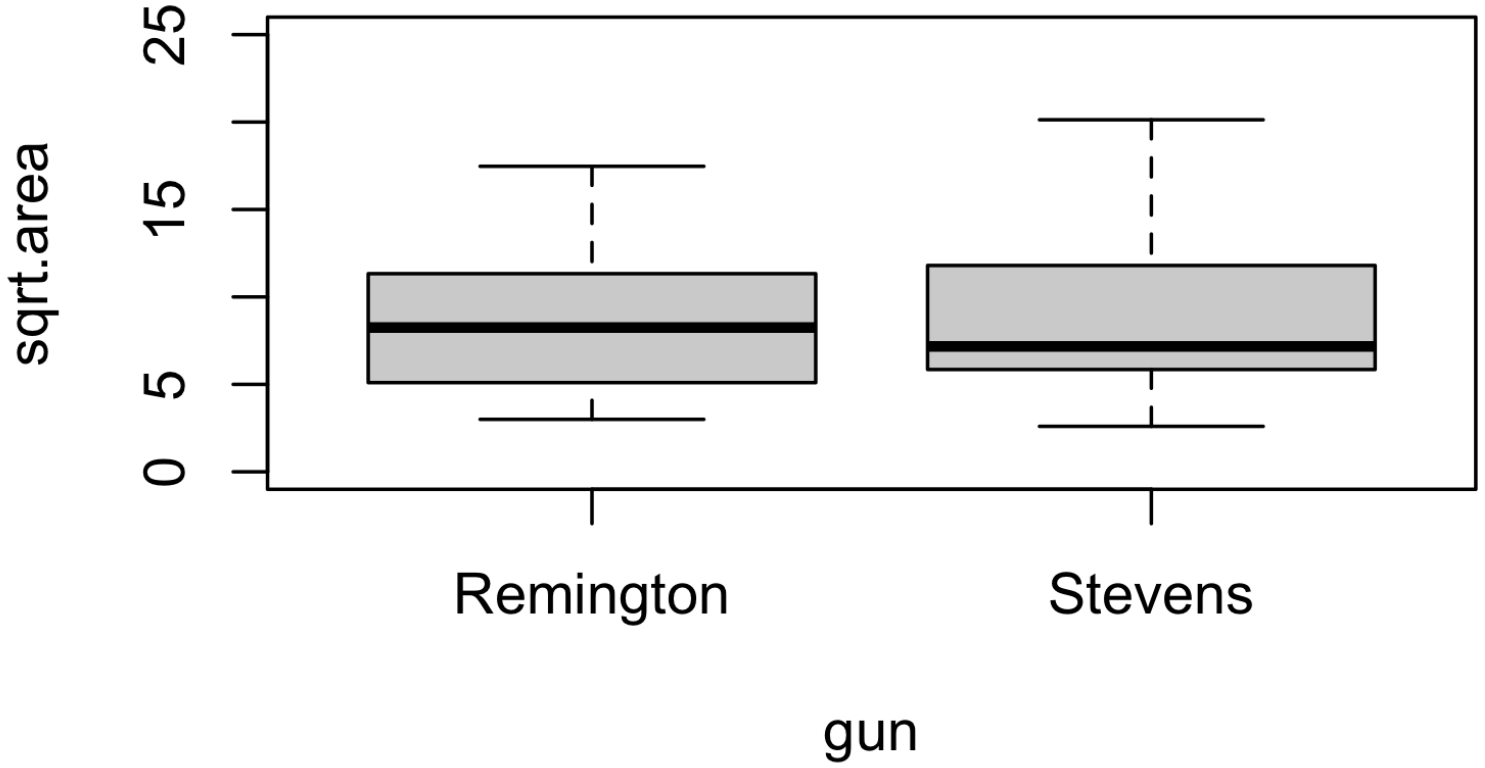
Skewness = 0.5
Kurtosis = 2.44

Computing descriptive statistics



Can parse your data on categories and create several “univariate numeric data” problems.

Computing descriptive statistics



Remington
Mean = 8.43
Median = 8.25

Variance = 18.16
St. Dev = 4.26

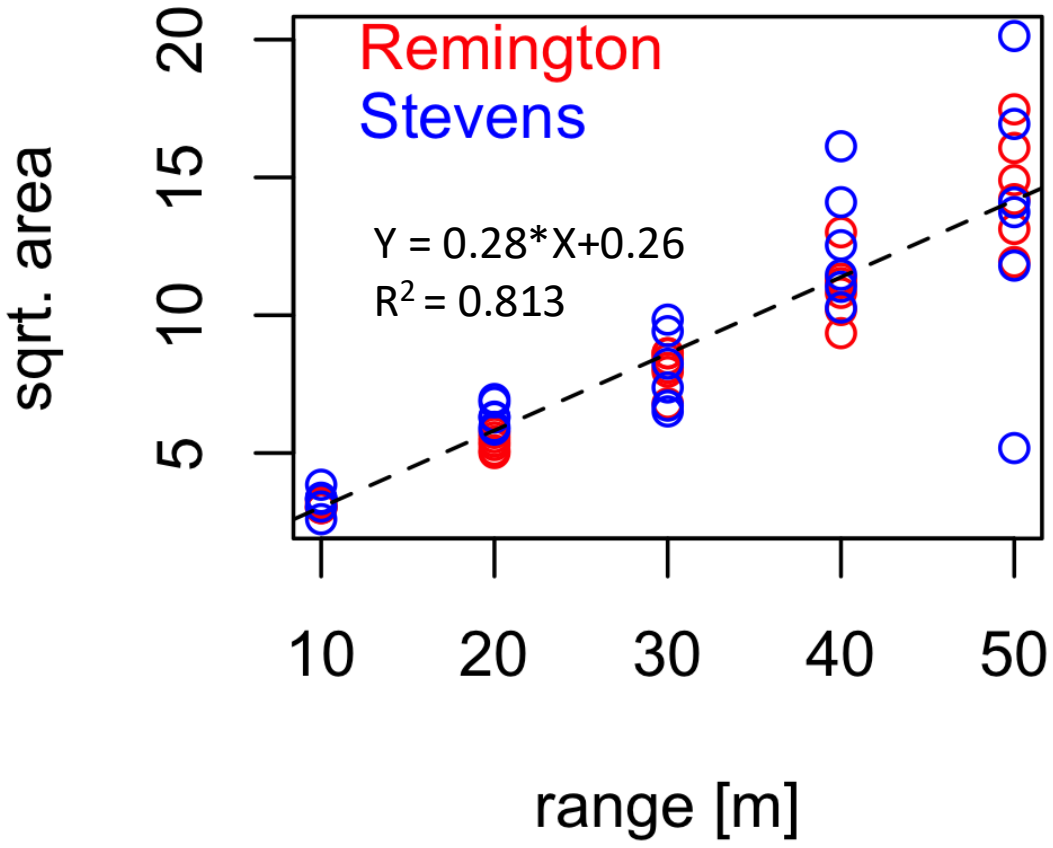
Stevens
Mean = 8.78
Median = 7.17

Variance = 21.17
St. Dev = 4.60

Computing descriptive statistics

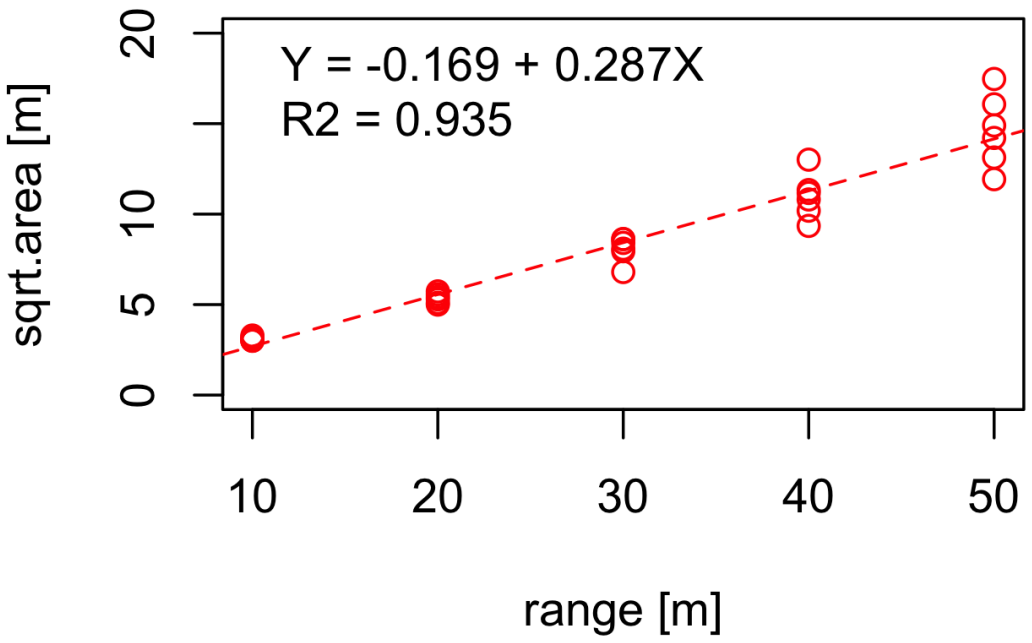
- What is the “central tendency”?
  - Correlation Coefficient
    - Strong vs weak, positive vs negative
  - Model Coefficients
    (e.g., slope and intercept of
    line of best fit)

- What is the “dispersion”?
  - Sum of squared variance

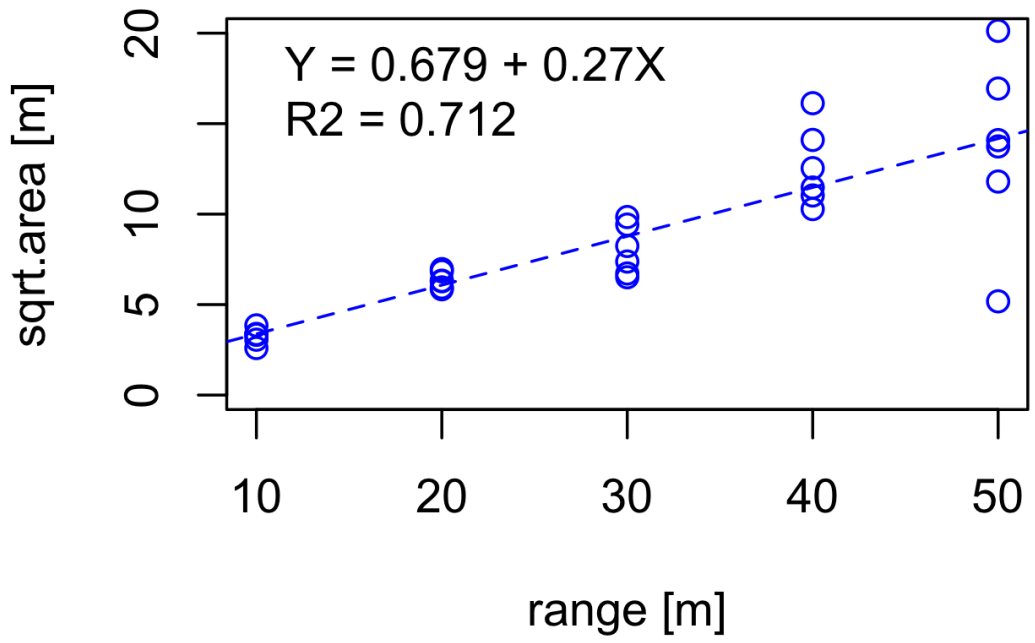


From a result to an interpretation

Remington



Stevens



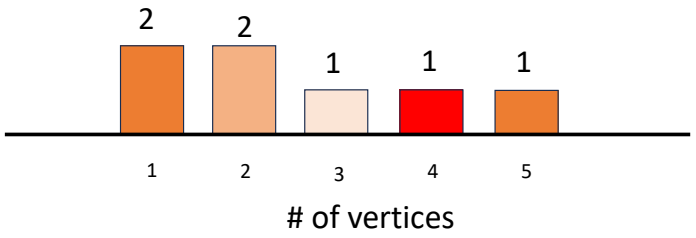
Inference: from sample to population

- Statistical inference is the process of inferring (or estimating) population parameters using sample statistics

  - Point Estimation
  - Interval Estimation
  - Hypothesis Testing

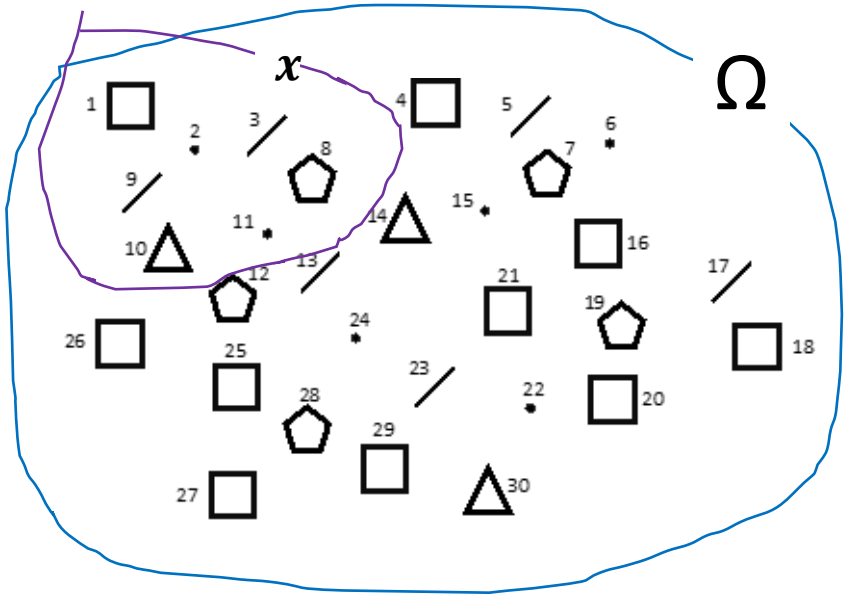
Inference: from sample to population

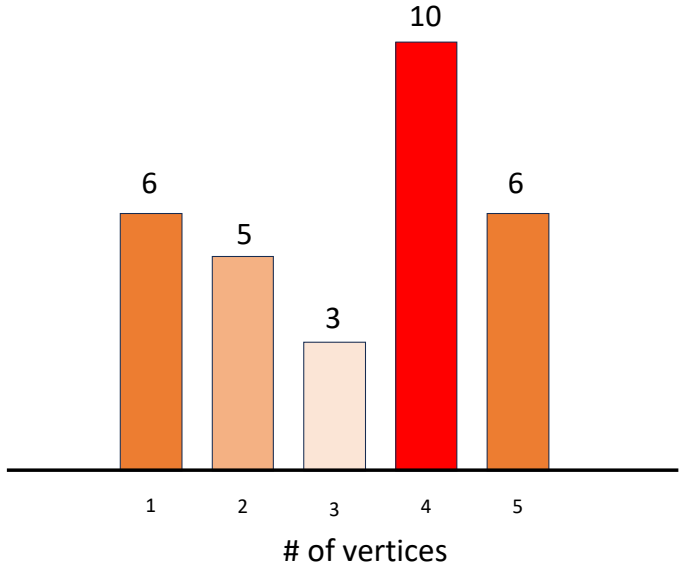
- Point estimation: The process of predicting a specific parameter value from a sample statistic



Sample Mean Number of Vertices

$\bar{x} = 2.57$



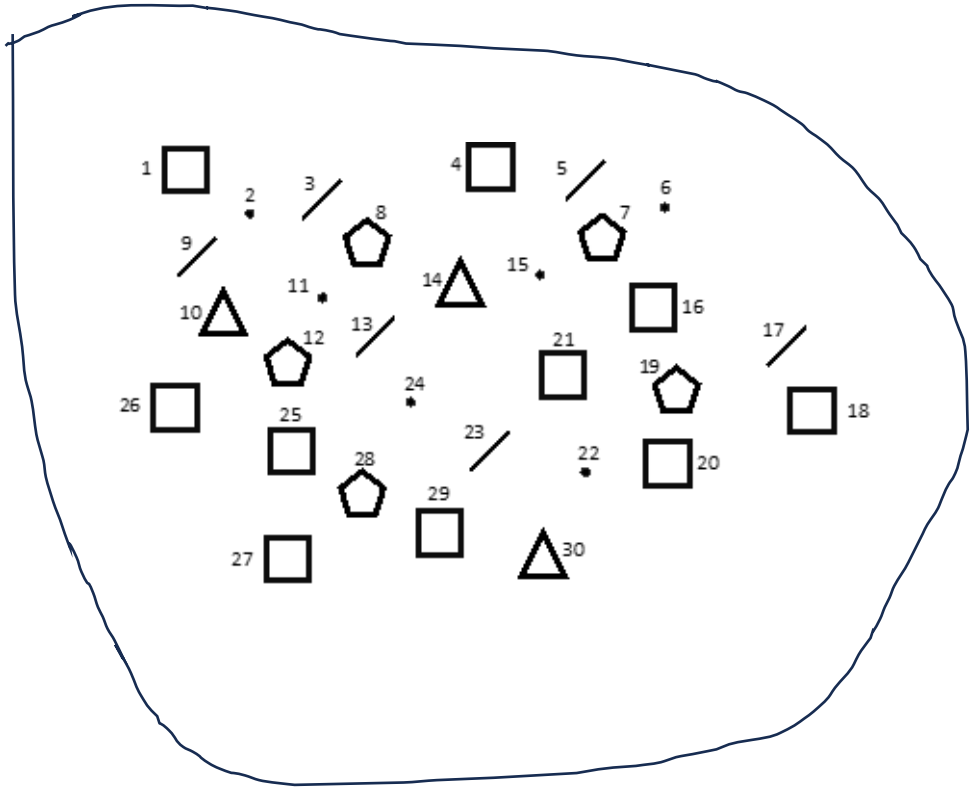


Mean Number of Vertices

$\bar{\Omega} = 3.167$

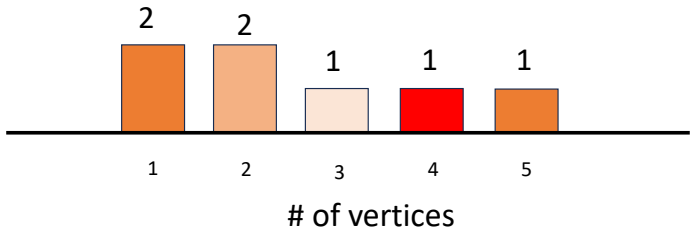
Some new concepts

- A sampling distribution is the distribution of a statistic, when computed for **randomly selected samples of the same size**

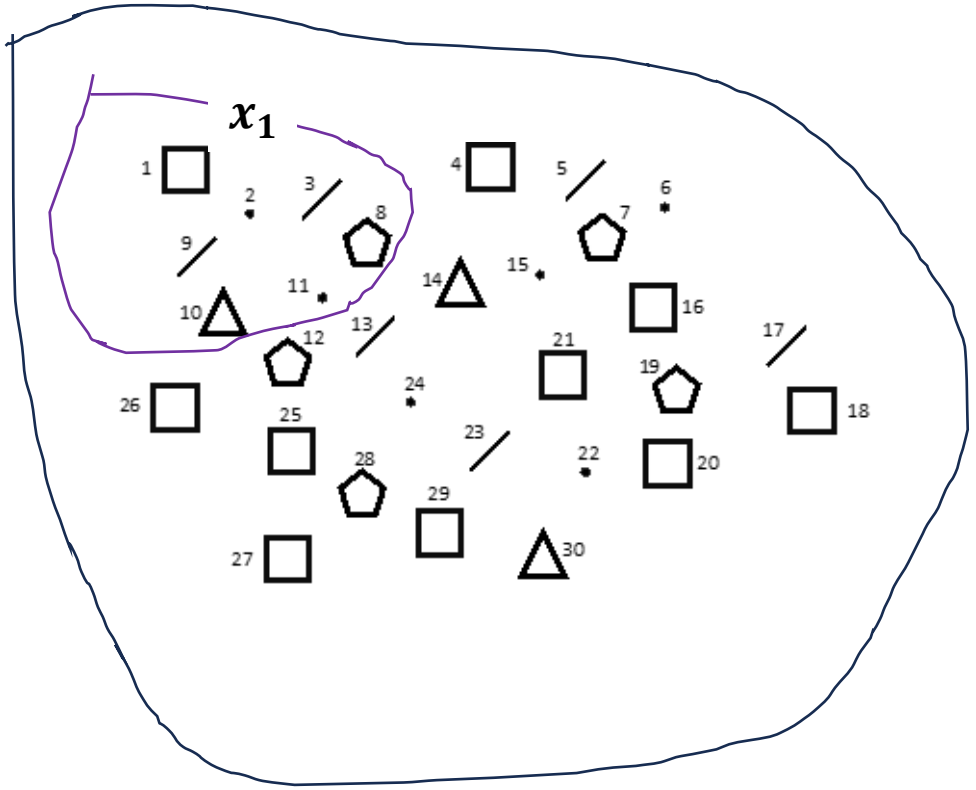


Some new concepts

- A sampling distribution is the distribution of a statistic, when computed for **randomly selected samples of the same size**

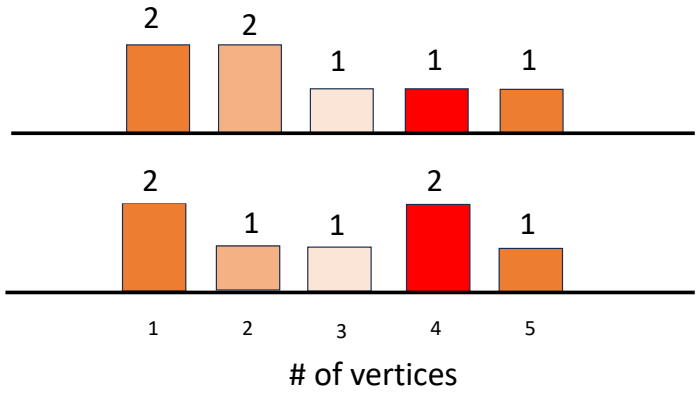


$\bar{x}_1 = 2.57$



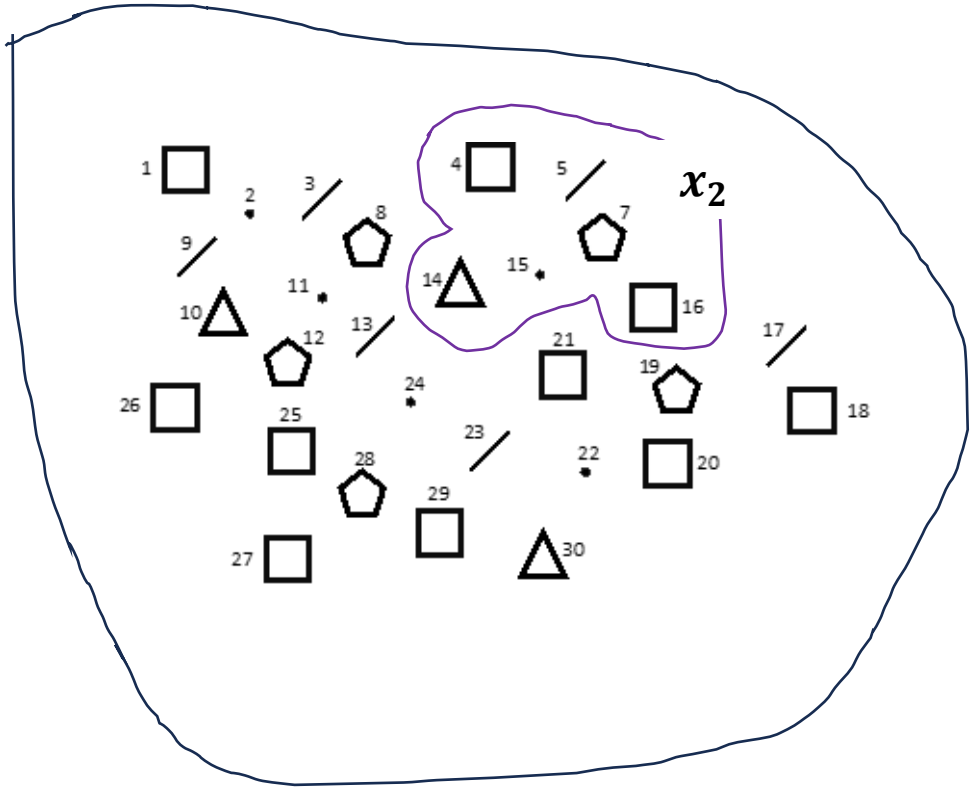
Some new concepts

- A sampling distribution is the distribution of a statistic, when computed for **randomly selected samples of the same size**



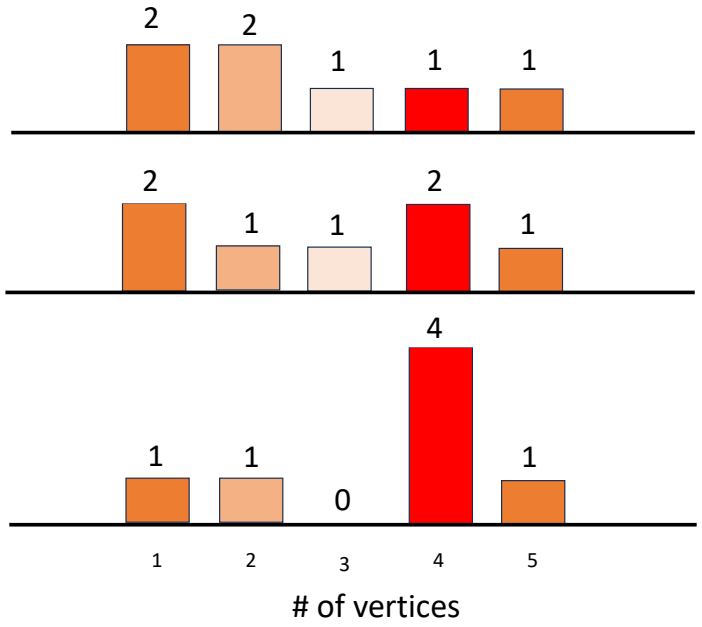
$\bar{x}_1 = 2.57$

$\bar{x}_2 = 2.86$



Some new concepts

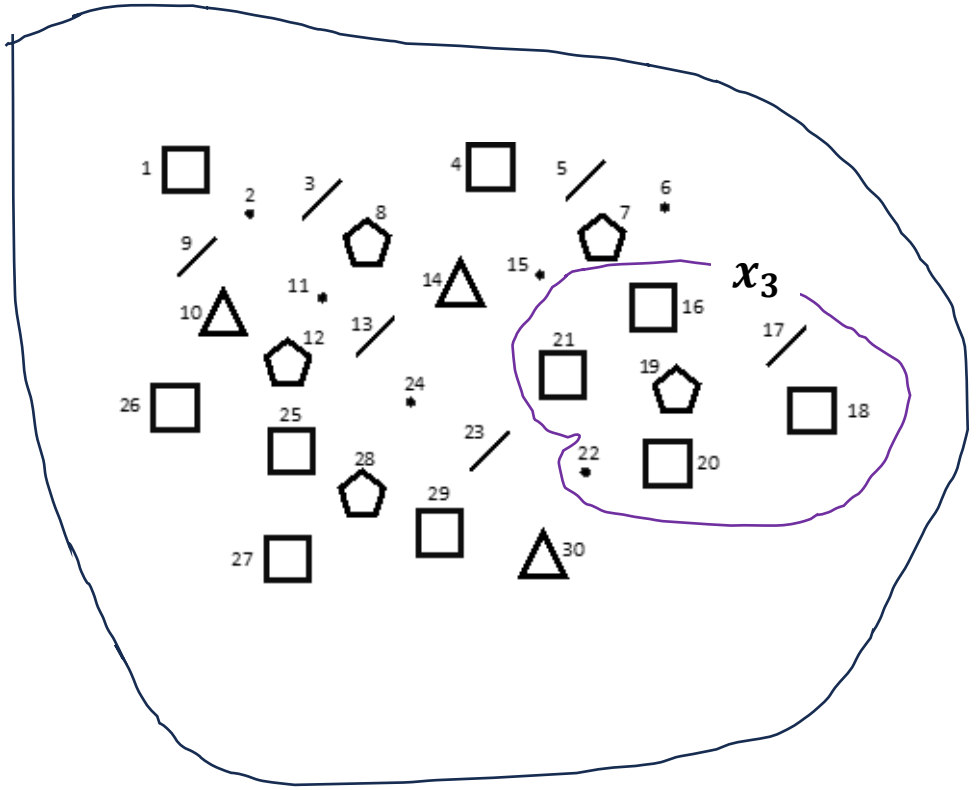
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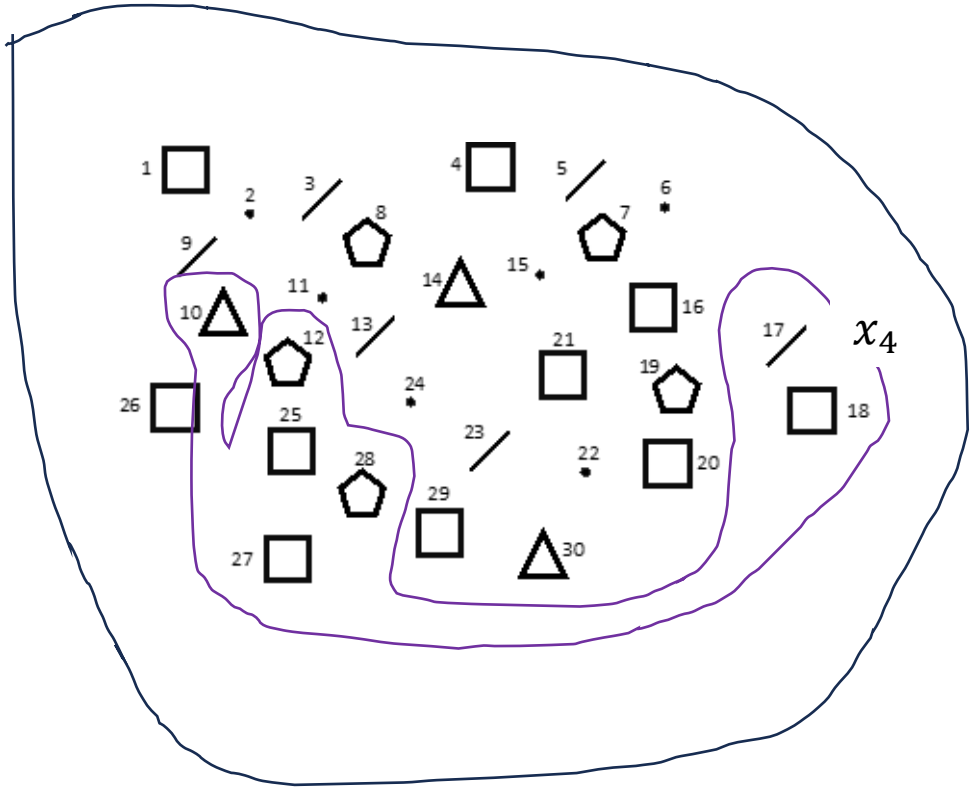
$\bar{x}_2 = 2.86$

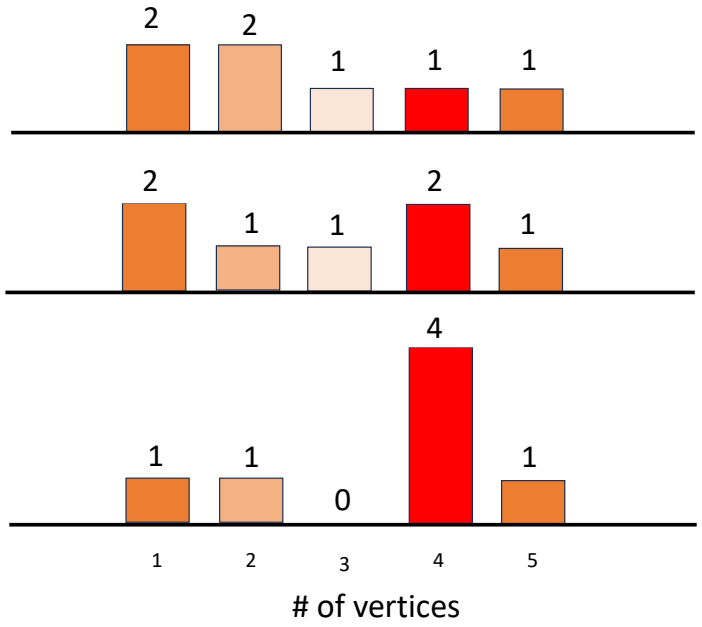
$\bar{x}_3 = 3.43$



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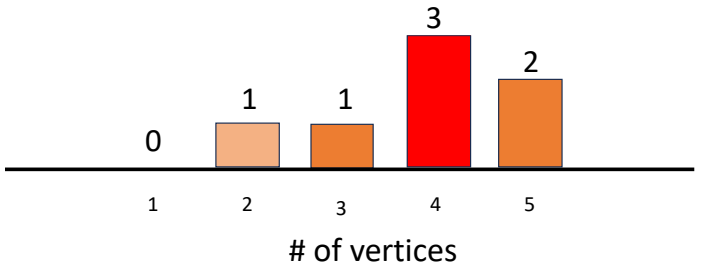


$\bar{x}_1 = 2.57$

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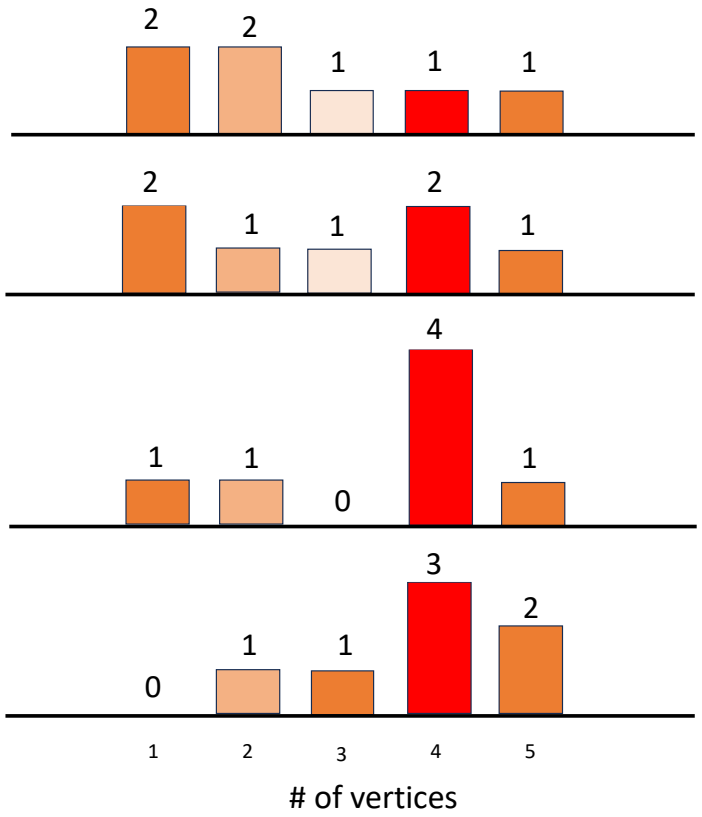
$\bar{x}_3 = 3.43$

$\bar{x}_4 = 3.86$



Some new concepts

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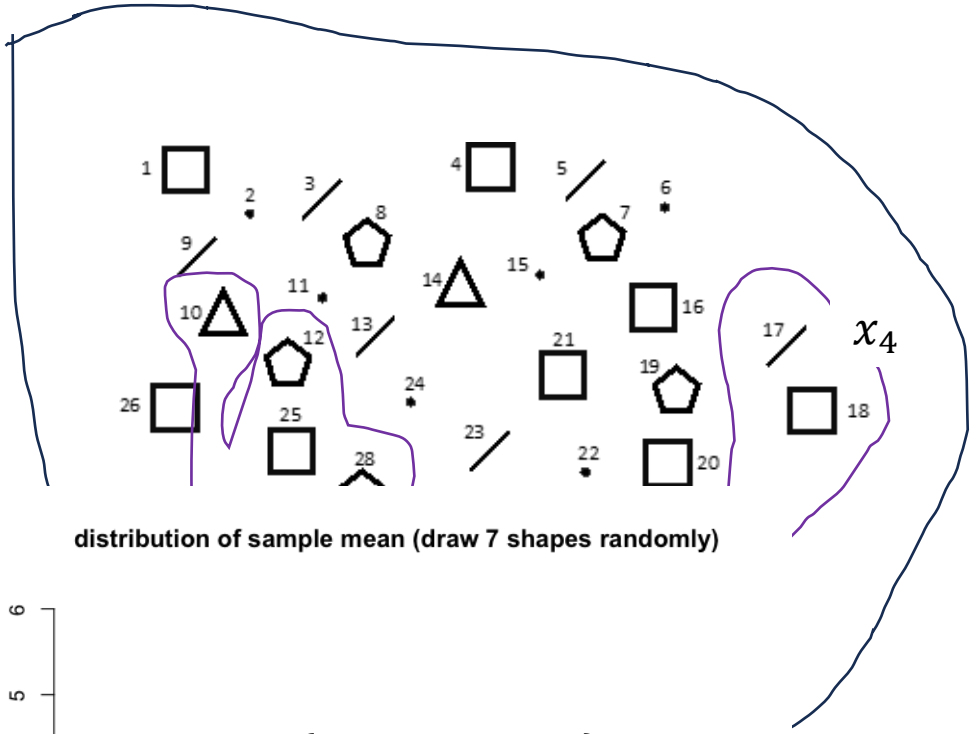


$\bar{x}_1 = 2.57$

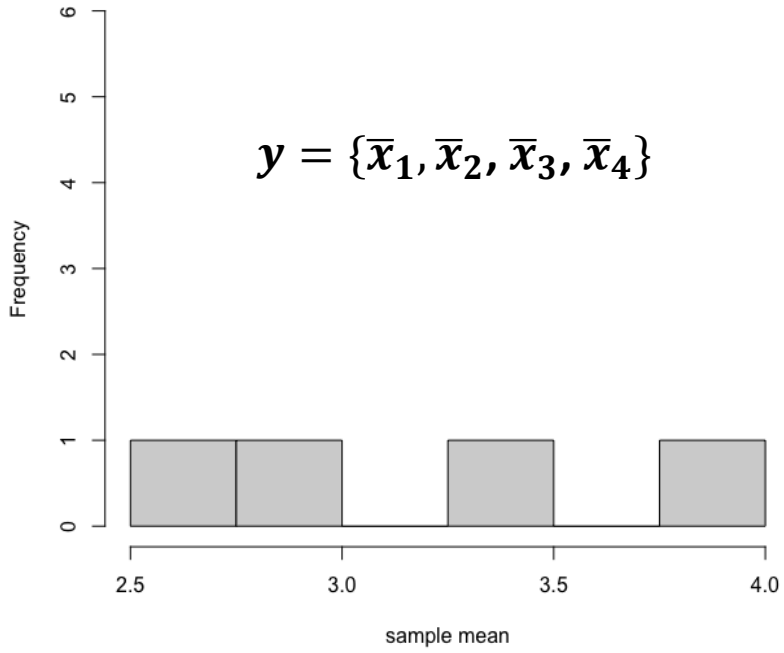
$\bar{x}_2 = 2.86$

$\bar{x}_3 = 3.43$

$\bar{x}_4 = 3.86$

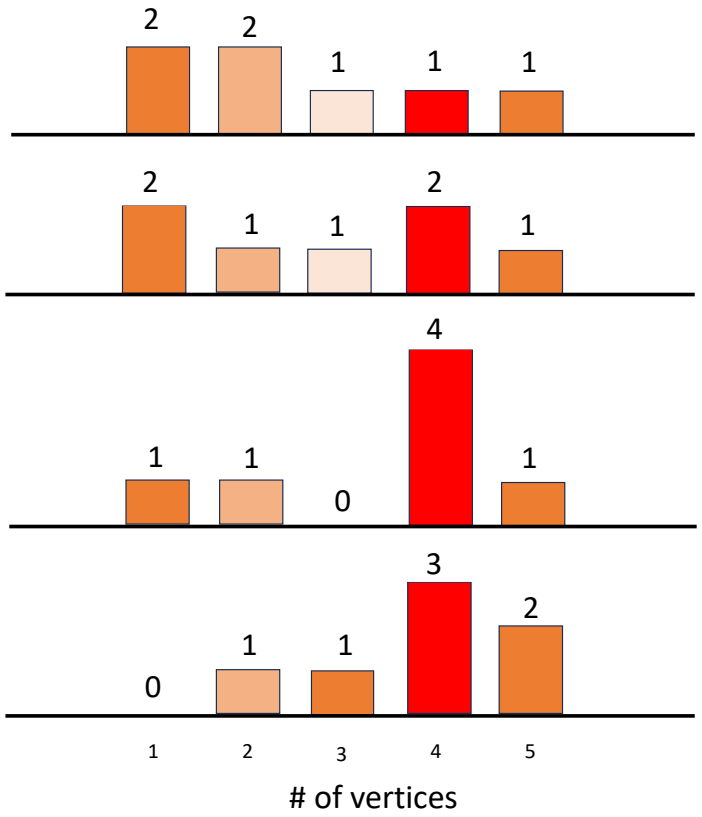


$y = \{\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4\}$



Some new concepts

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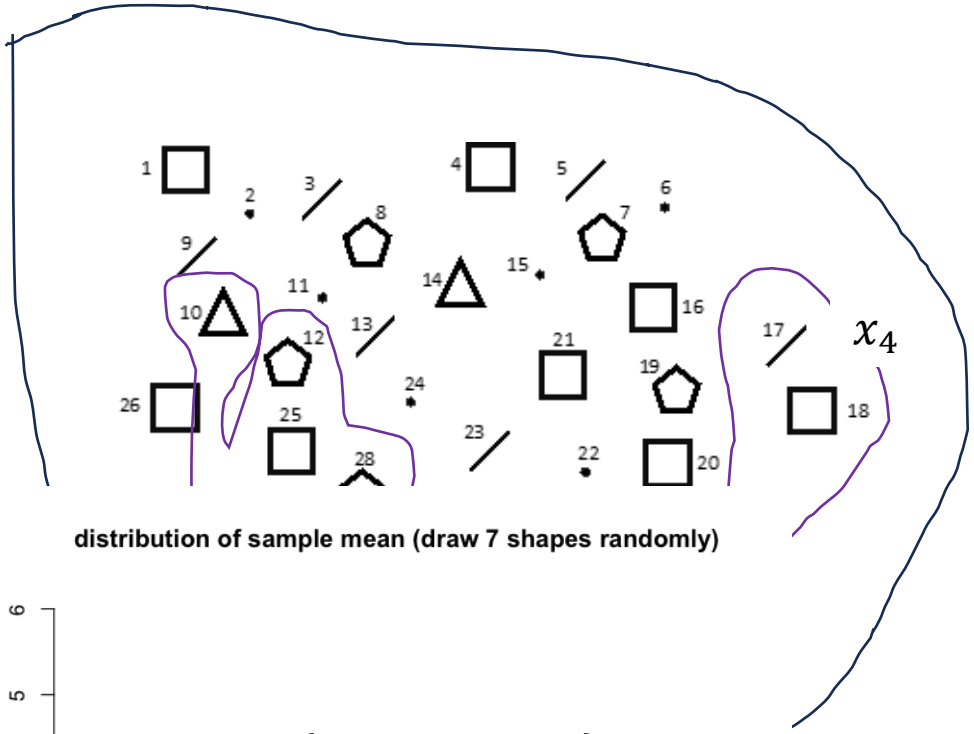


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$\bar{x}_2 = 2.86$

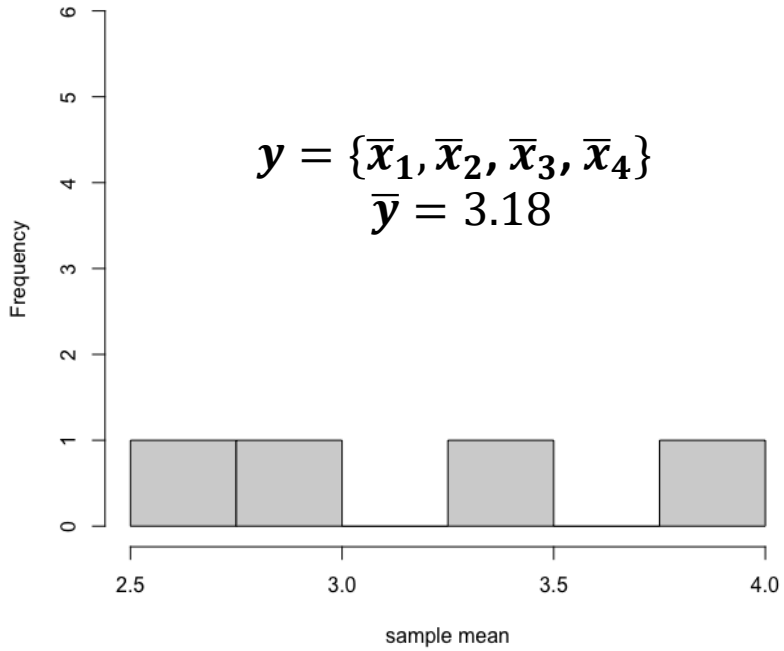
$\bar{x}_3 = 3.43$

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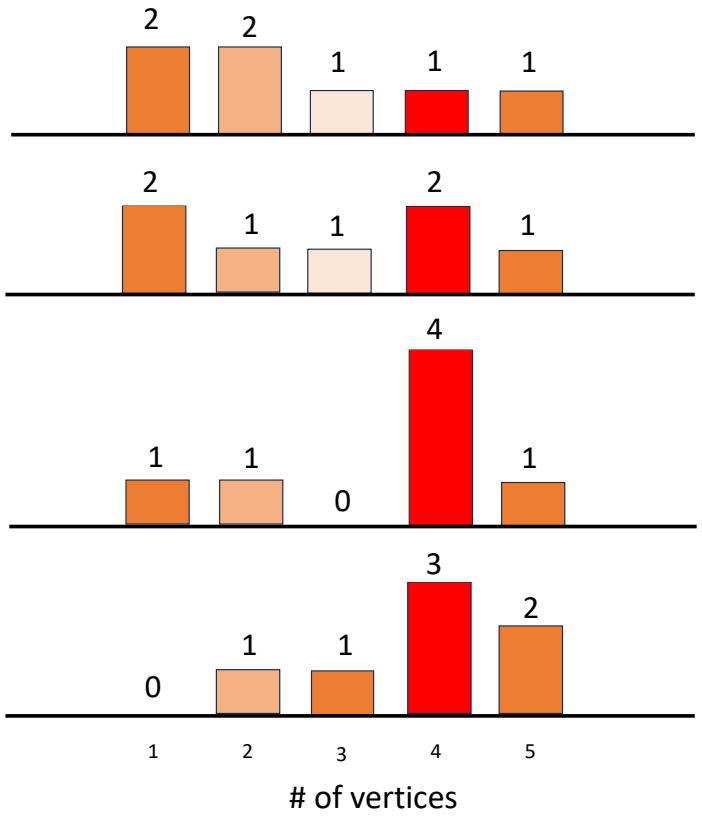
$y = \{\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4\}$

$\bar{y} = 3.18$



Some new concepts

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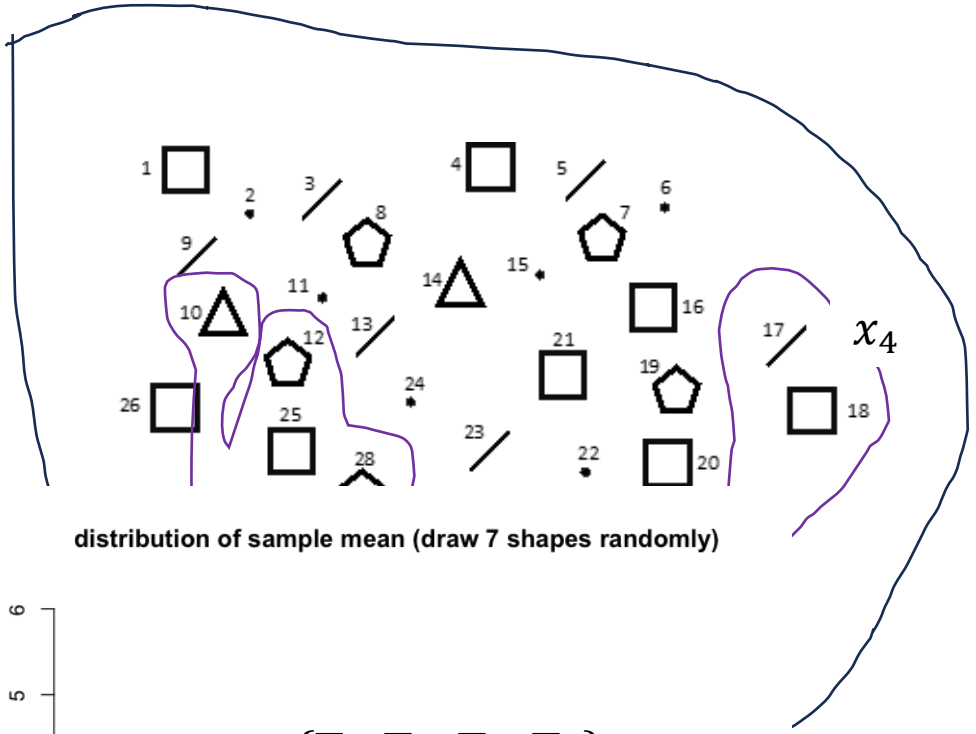


$\bar{x}_1 = 2.57$

$\bar{x}_2 = 2.86$

$\bar{x}_3 = 3.43$

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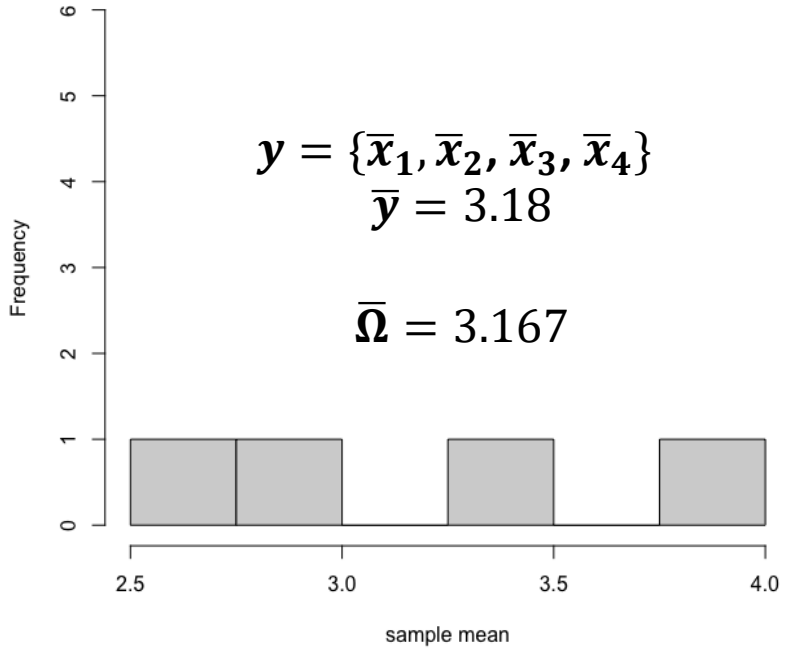


distribution of sample mean (draw 7 shapes randomly)

$y = \{\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4\}$

$\bar{y} = 3.18$

$\bar{\Omega} = 3.167$



Inference: from sample to population

- Statistical inference is the process of inferring (or estimating) population parameters using sample statistics

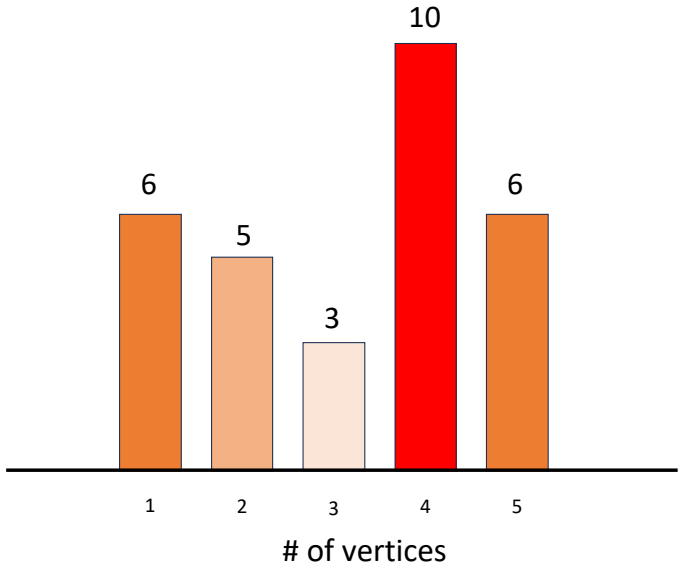
    - Point Estimation
    - **Interval Estimation**
    - Hypothesis Testing

Interval estimation

- **Interval estimation:** The process of predicting a range (interval) within which a parameter value exists.

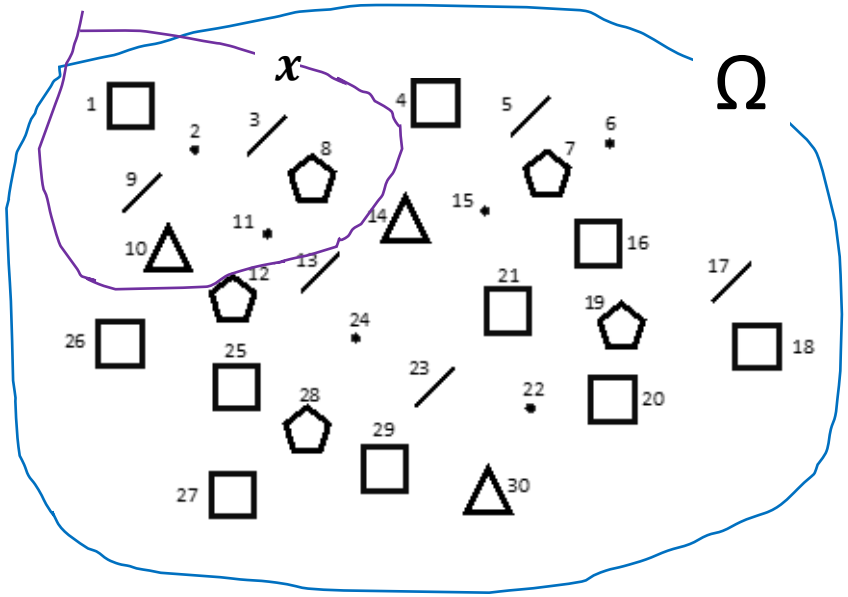
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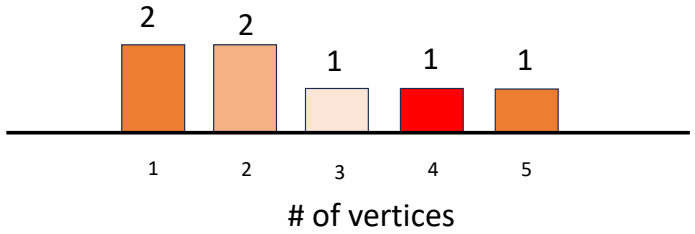
- **Interval estimation:** The process of predicting a range (interval) within which a parameter value exists.



Mean Number of Vertices

$\bar{\boldsymbol{\Omega}} = 3.167$



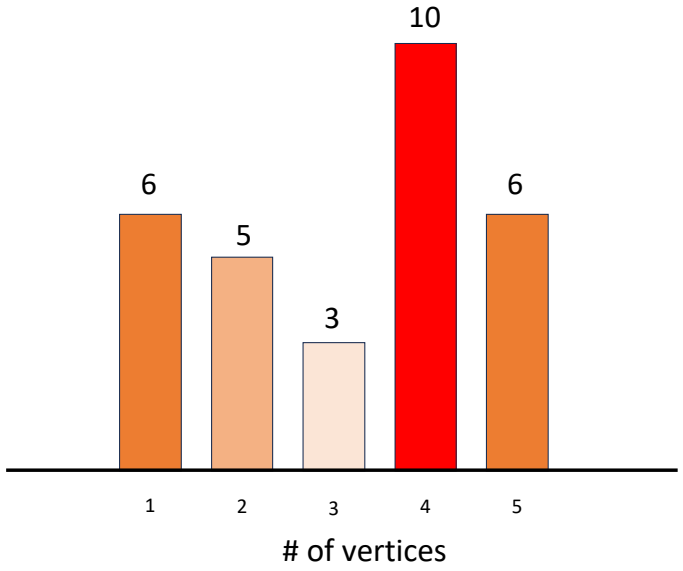


Range of values in $\boldsymbol{x}$:

$\boldsymbol{R} = [1 - 5]$

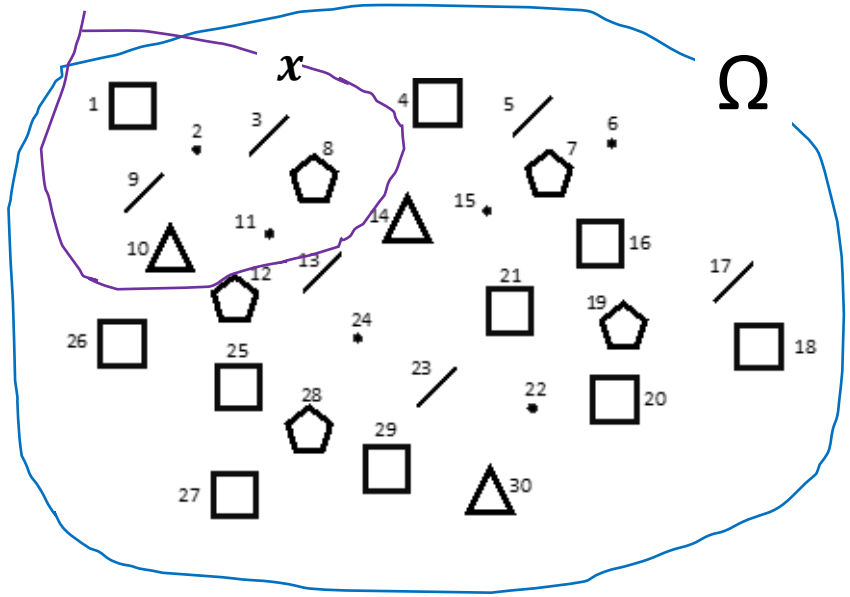
Interval estimation

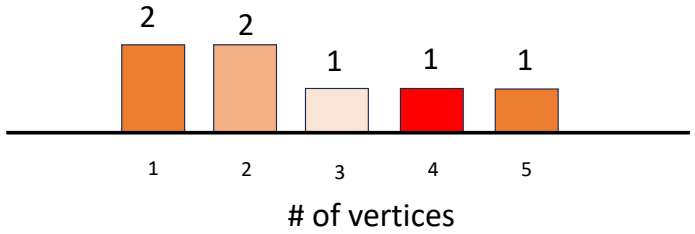
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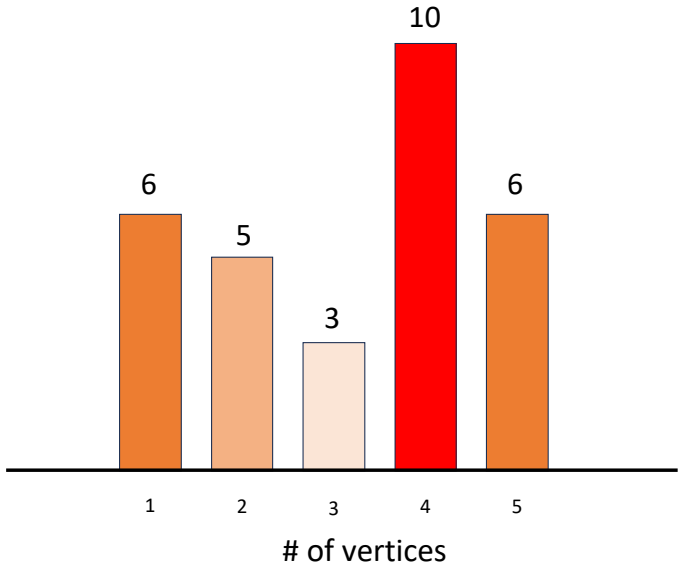
Range of values in x:

$R = [1 - 5]$

This range does encapsulate the parameter value—but it's not very precise.

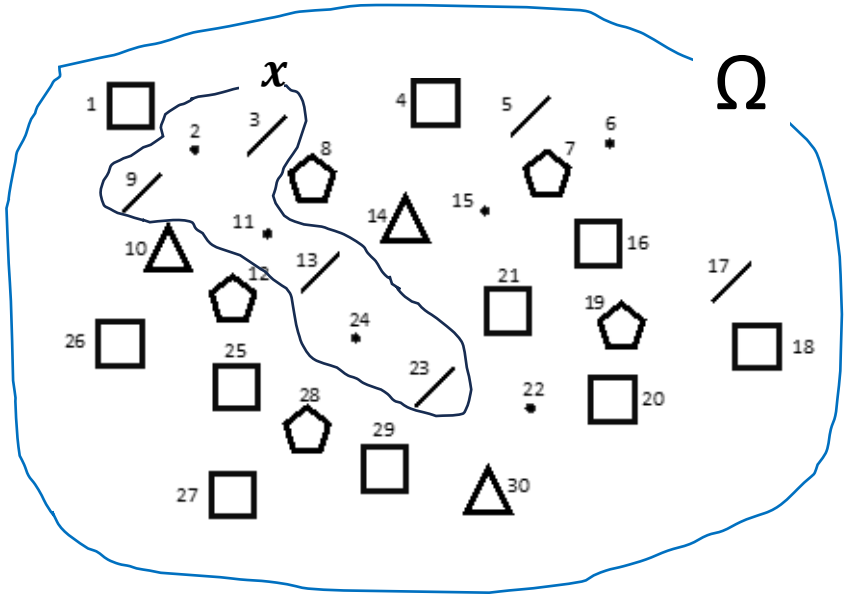
Interval estimation

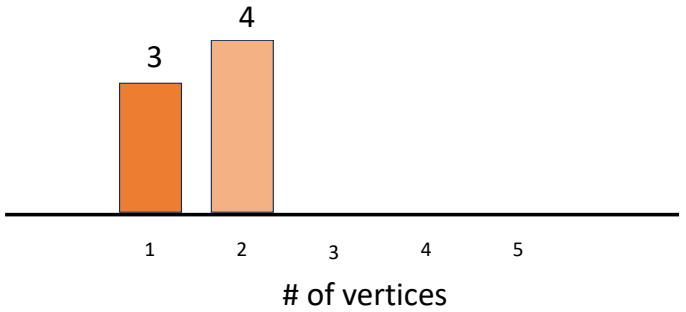
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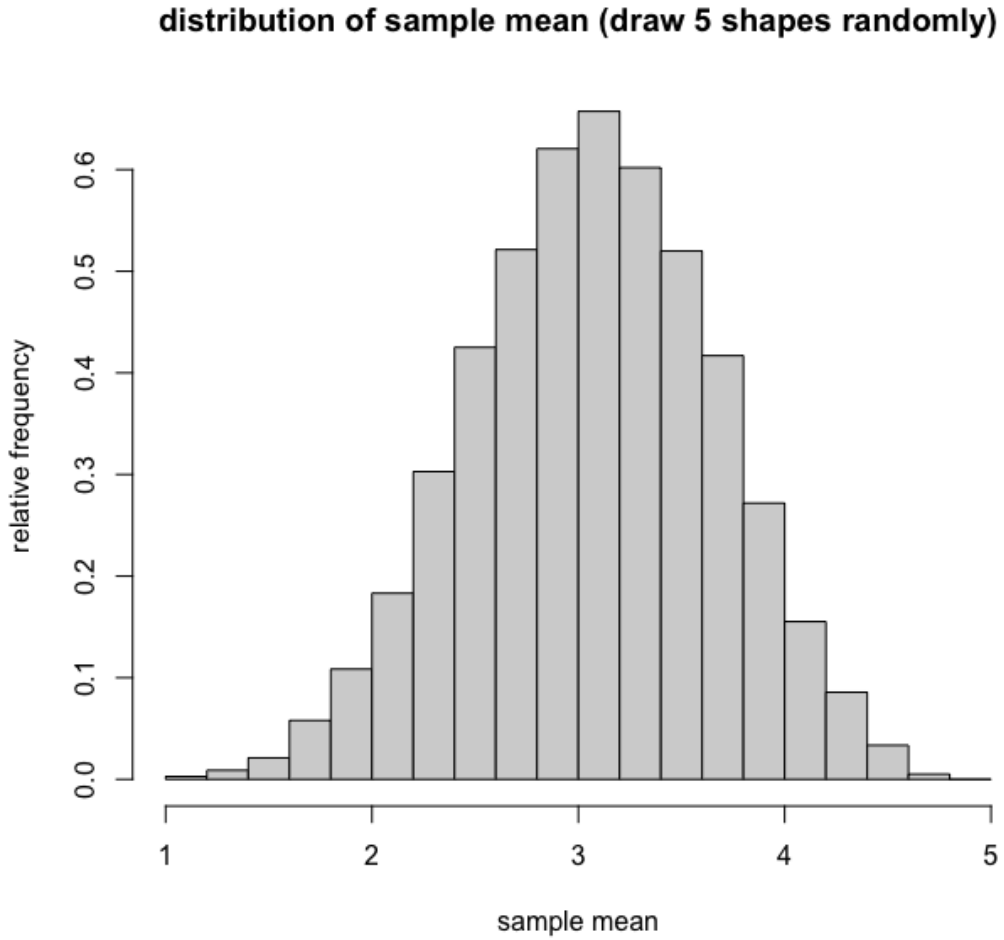
Range of values in $\boldsymbol{x}$:

$\boldsymbol{R} = [1 - 2]$

Also, making an inference from a single random selection of data could fail – by chance.

Interval estimation

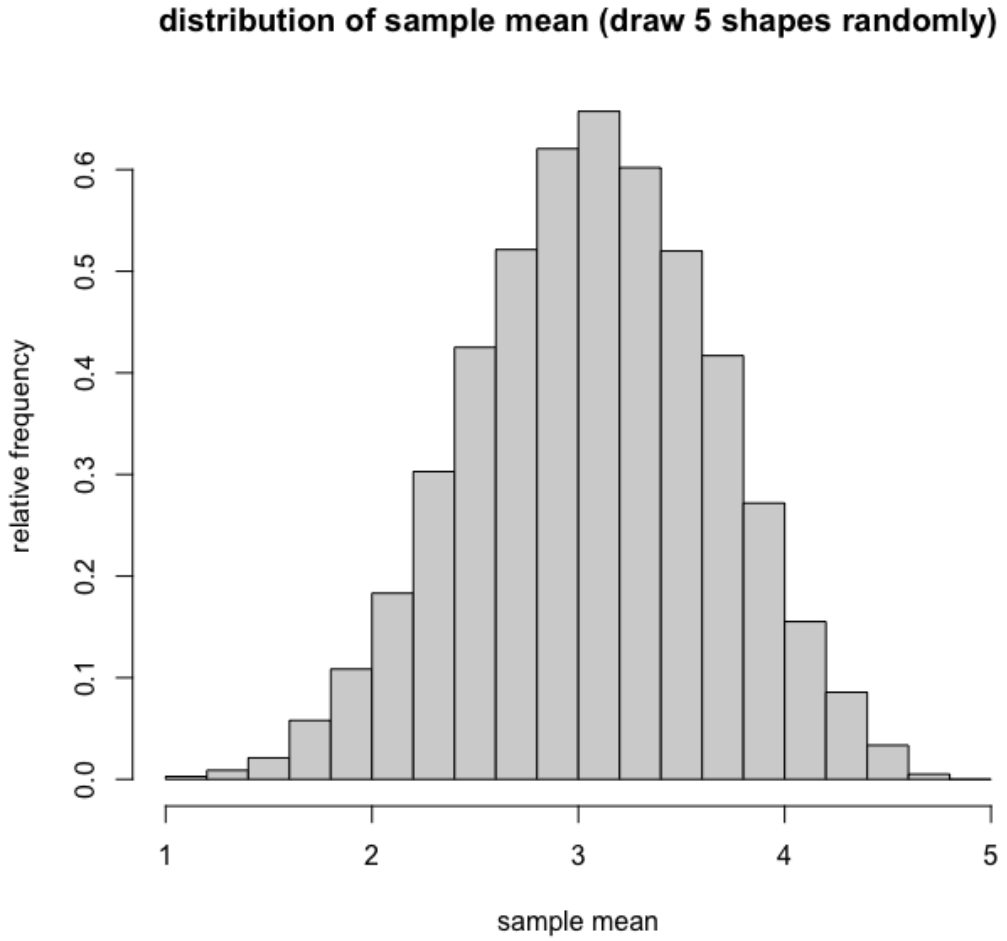
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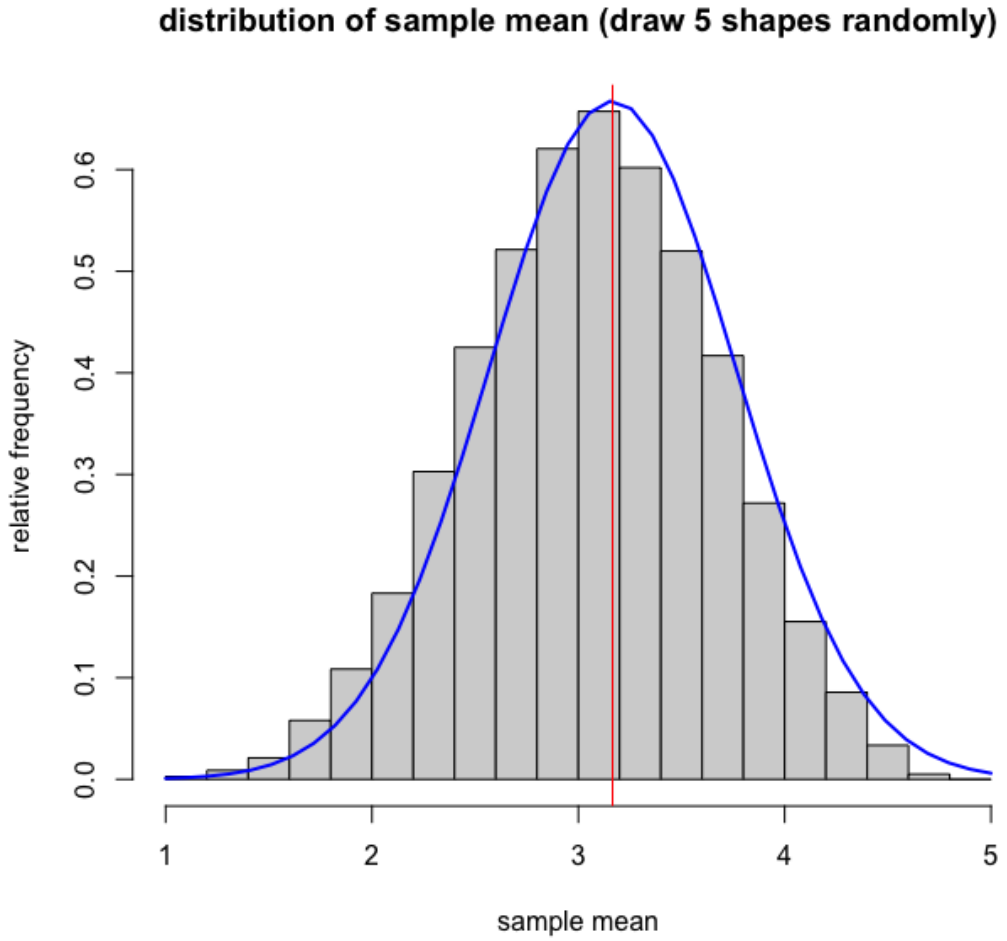
- From **central limit theorem**, we know that *all sampling distributions tend to a normal distribution* as the number of samples approaches infinity



Interval estimation

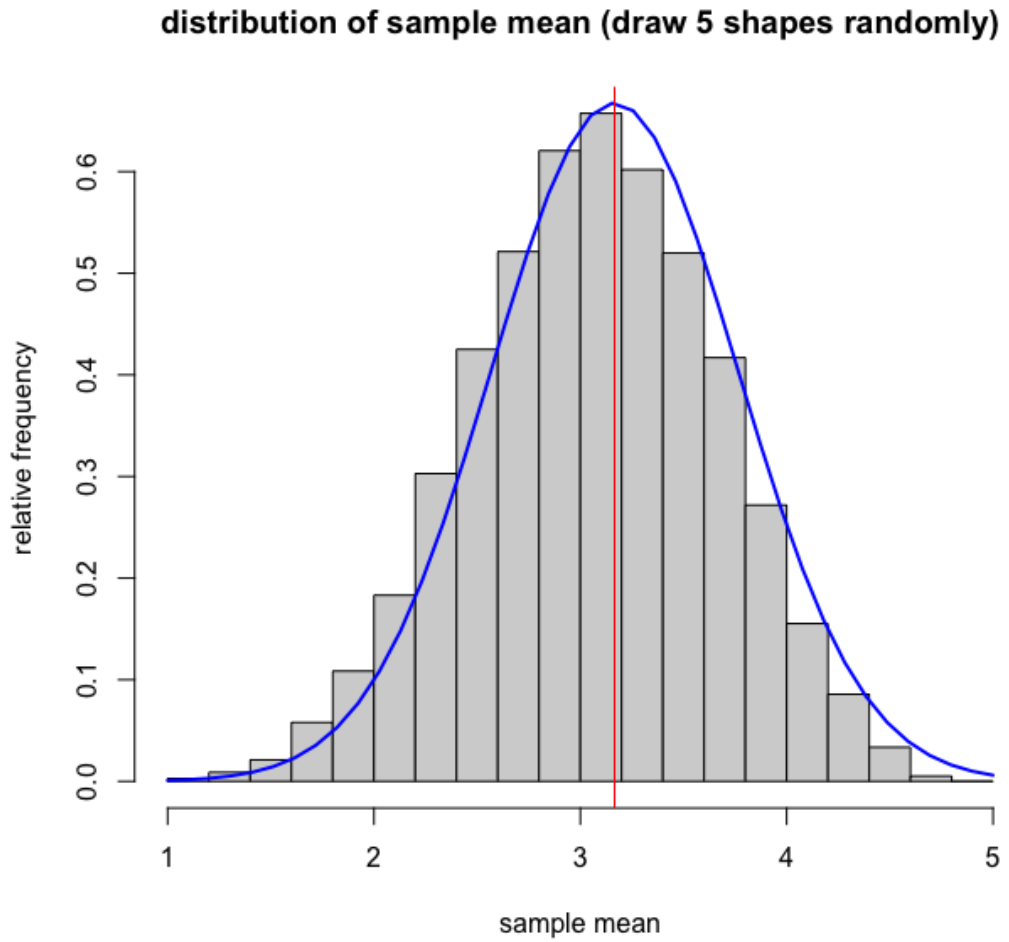
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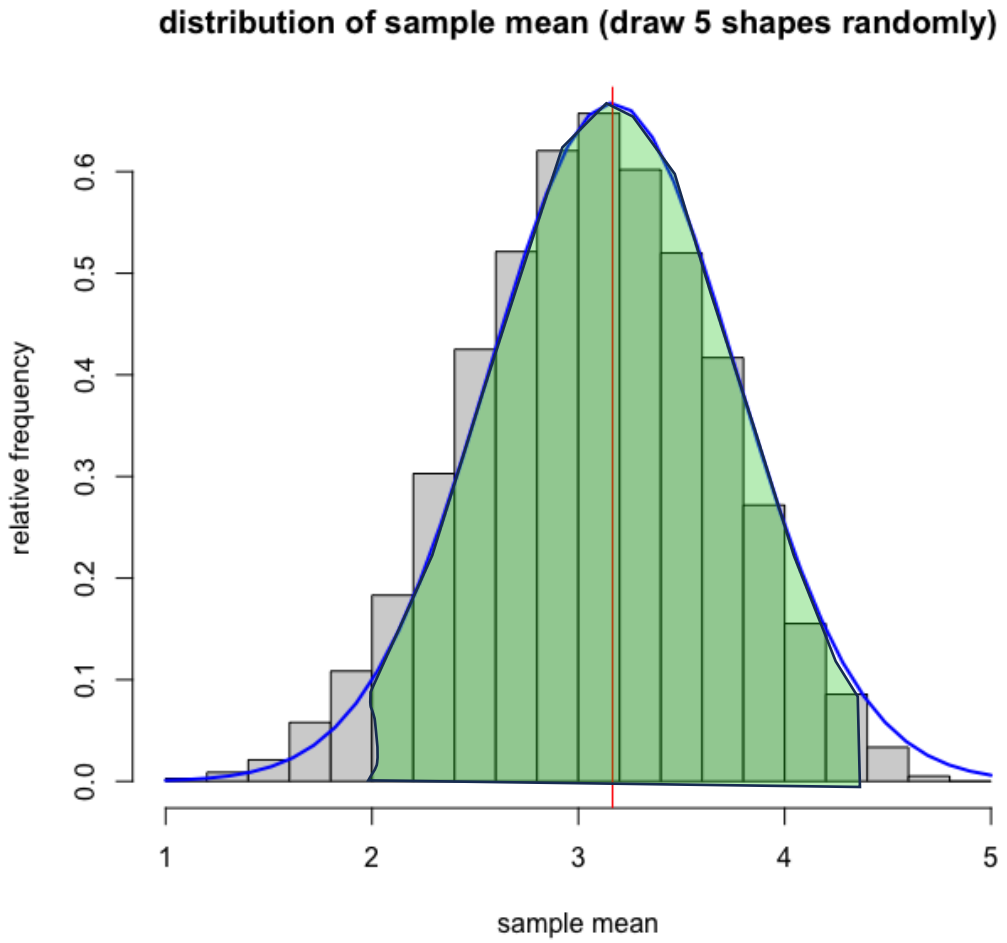
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Interval estimation

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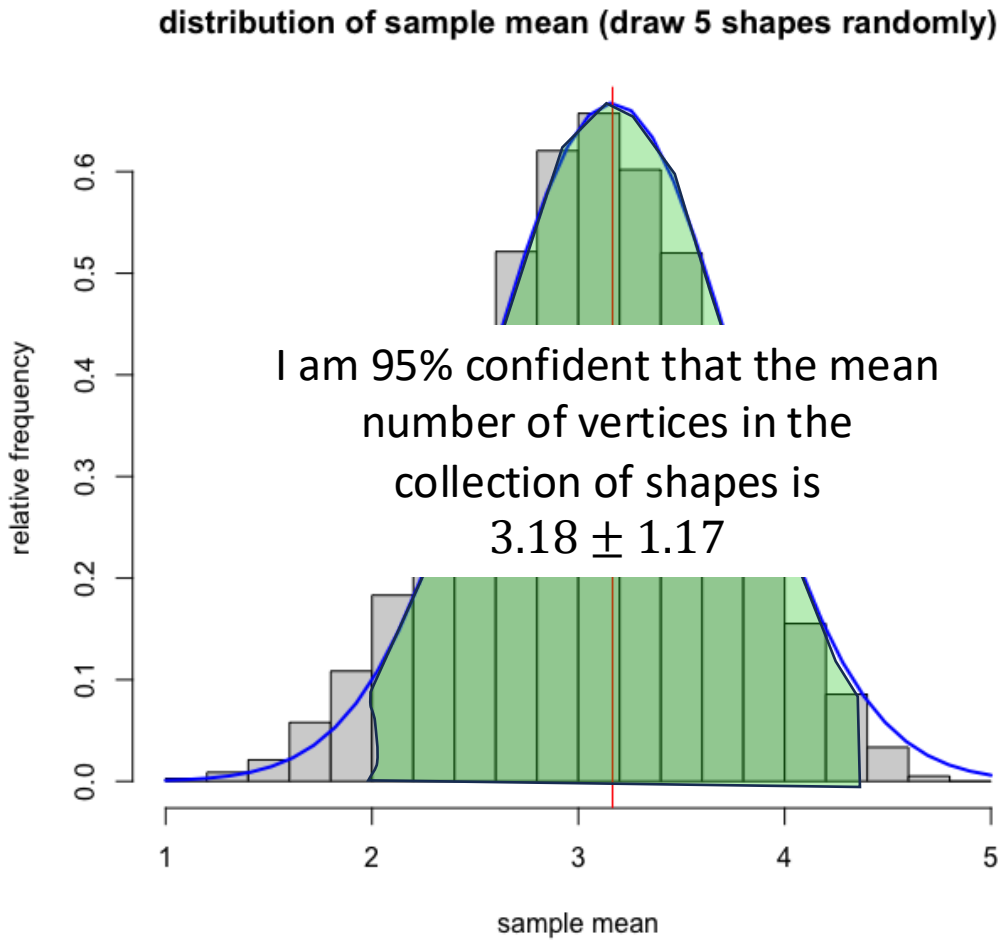
- For a variable that follows a normal distribution, we know that **95% of the data falls within ± 1.96 *se* of the mean value** – refer to this as a confidence interval



Interval estimation

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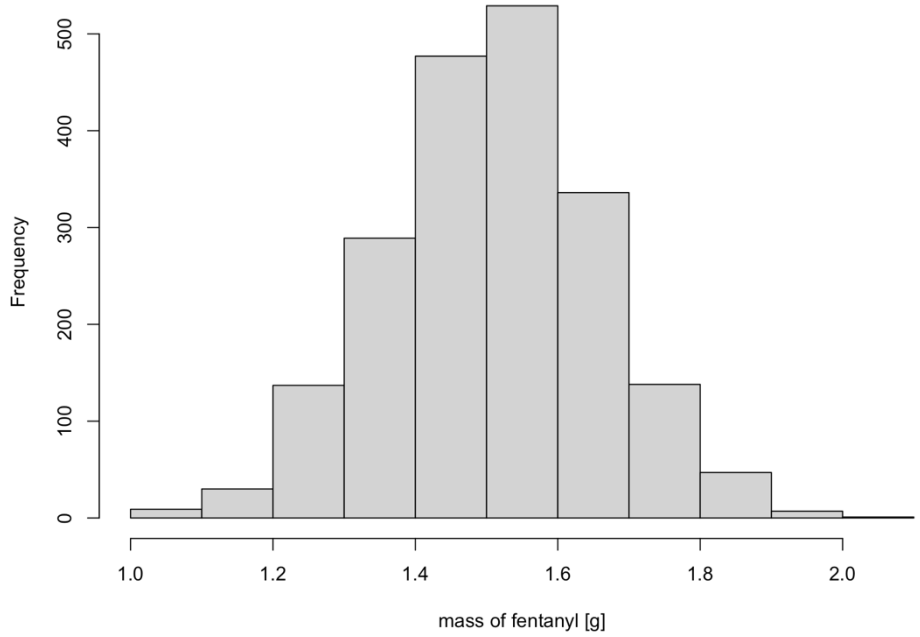


Working through an example

SCENARIO: You're given the fentanyl mass measurements for 2000 pills (see histogram to the right). The mean mass for your set is 1.51 mg and the standard deviation for the set is 0.15 mg.

Given this information, what is your best estimate for the mass of fentanyl in a new pill?

Answer 1: 1.51 mg



Working through an example

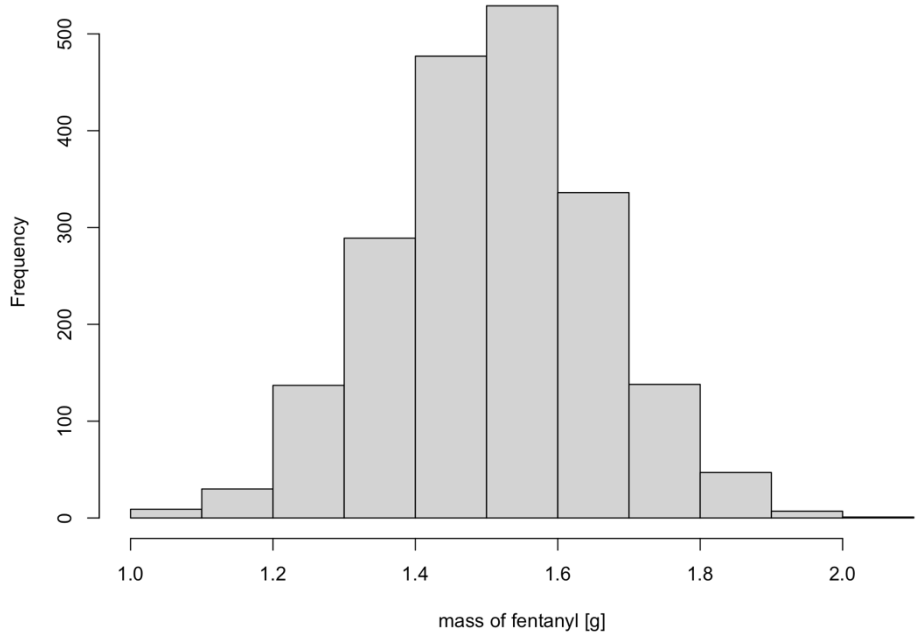
Answer 2: Assuming that the mass measurements are normally distributed, we know that 95% of the data is within approximately 2 standard errors of the mean.

Mean = 1.51; Standard Deviation = 0.15; n = 2000

Standard Error = 0.15/sqrt(2000) = 0.003

95% CI = 2 x 0.003 = 0.006

Therefore, 1.51±0.01 mg (1.50 – 1.52 mg) with 95% confidence



Inference: from sample to population

- Statistical inference is the process of inferring (or estimating) population parameters using sample statistics
    - Point Estimation: Using sample statistics to predict/estimate a specific parameter value.
    - Interval Estimation: Using sample statistics to predict/estimate a range (interval) the parameter value lives in.
    - **Hypothesis Testing:** Using sample statistics to evaluate an assumed parameter value.

Key Terms

- **Hypothesis:** A premise or claim for the value of a parameter.
- **Null Hypothesis (H_0)** – The accepted hypothesis. *(no change)*
- **Alternative Hypothesis (H_a)** – The competing hypothesis. *(some change)*
- **Note:** Null and alternative hypothesis must be mutually exclusive (both premises cannot be simultaneously true)

Example

Based on historic seizure data, the average amount of fentanyl found in a typical illicitly distributed pills is 0.5 mg. Researchers believe that estimate of the mean is not representative of the current drug market. Outline the null and alternative hypotheses that they should test.

$$H_0: \mu = 0.5$$

$$H_a: \mu \neq 0.5$$

Key Terms

- **Hypothesis Test:** A statistical procedure for evaluating two competing hypotheses using sample evidence (statistics), the results of which allow us to:
  1. Reject the null hypothesis, or
  2. Fail to reject the null hypothesis

- **Significance Level (α):** A user-prescribed value (between 0 and 1) for deciding whether the sample evidence is sufficient (or insufficient) for rejecting the null hypothesis.

  Compliment of a confidence level, i.e., ($\alpha = 1 - C$)

Key Terms

- **P-value:** The probability that you would observe the sample evidence *if the null hypothesis were true*.
  - If p-value is less than α, the hypothesis test rejects the null hypothesis.
  - If p-value is greater than α, the hypothesis test fails to rejects the null hypothesis
  - If p-value is less than α, the observed “effect size” (difference between parameter values) is said to be significant

Key Terms

- Type 1 Error (false positive conclusion):
  - Test incorrectly rejects H_0
  - To minimize Type 1 errors, you want to decrease α

- Type 2 Error (false negative conclusion):
  - Test incorrectly fails to reject H_0
  - To minimize Type 2 errors, you want to increase “statistical power”

How all hypothesis tests work

1. Set significance level (α) for hypothesis test

2. Use sample statistics and assumed parameter values to compute a **test statistic** (sample evidence)

3. Determine the p-value for **test statistic** using probability distribution implied by the null hypothesis

4. Compare p-value to significance level to make inference

Which hypothesis test should you use?

- Depends on several factors
  - Data type (continuous vs ordinal vs categorical)
  - Population parameter being estimated / question being asked

- Important hypothesis tests:
  - T-tests
  - ANOVA
  - Non-parametric versions

Example 1

Scenario: The average concentration of THC in seized marijuana is 4%. In a recent seizure of 20-1g bags, the concentration of THC is as described on the right.

How different is the average concentration of THC in the recent seizure from past seizures? Is this difference significant?

Obs.	Conc.	Obs.	Conc.
1	6.89	11	7.22
2	2.33	12	8.32
3	7.82	13	3.41
4	3.91	14	5.63
5	6.02	15	8.23
6	7.54	16	9.21
7	6.02	17	3.87
8	6.37	18	7.45
9	6.17	19	5.35
10	5.21	20	5.41

(Student's) T-test

- A hypothesis test used to evaluate whether there is a significant difference (in mean) between **two** groups
  - One-sample T-test: When comparing a sample to a population
  - Two-sample T-test: When comparing two independent samples
  - Paired T-test: When comparing samples that are not independent

- Requirements: Data is continuous, normally distributed
- Test Statistic: T-statistic ; Test Distribution: T-distribution
- Non-parametric alternative: Wilcoxon

Solution 1

Finding the difference between averages is easy: The reported average is 4% whereas the sample average is 6.14%. Therefore, the difference is 2.14%.

To determine the significance of this difference, we can turn to a t-test (one sample t-test)

	A	B	C	D
1	6.89		sample mean	6.136
2	2.33		sample standard deviation	1.789605661
3	7.82		n	20
4	3.91		mu	4
5	6.02		t stat	5.337758261
6	7.54		t critical at 0.05	3.7599E-05
7	6.02			
8	6.37			
9	6.17			
10	5.21			
11	7.22			
12	8.32			
13	3.41			
14	5.63			
15	8.23			
16	9.21			
17	3.87			
18	7.45			
19	5.35			
20	5.75			
21				
22				

D5=(D1-D4)/(D2/SQRT(D3))

D6=T.DIST.2T(D5,D3-1)

Solution 1

Finding the difference between averages is easy: The reported average is 4% whereas the sample average is 6.14%. Therefore, the difference is 2.14%.

To determine the significance of this difference, we can turn to a t-test (one sample t-test)

```
> x = c(6.89, 2.33, 7.82, 3.91, 6.02, 7.54, 6.02, 6.37,
6.17, 5.21, 7.22, 8.32, 3.41, 5.63, 8.23, 9.21, 3.87,
7.45, 5.35, 5.75)
> mu = 4
> t.test(x,mu=mu)

        One Sample t-test

data:  x
t = 5.3378, df = 19, p-value = 3.76e-05
alternative hypothesis: true mean is not equal to 4
95 percent confidence interval:
 5.298439 6.973561
sample estimates:
mean of x
    6.136
```

Example 2

Scenario: Your lab measures the percentage THC in three different strains of marijuana from 5 different locations. The measurements are summarized in the data matrix to the right.

Are there significant differences due to location or strain?

location	strain	percentage
A	1	3.85618697
A	2	4.06354438
A	2	3.99682508
B	1	3.94848585
B	2	4.09798208
B	3	4.10475134
C	1	3.94827211
C	2	4.05966656
C	3	4.02955828
D	1	3.7740048
D	2	4.11053965
D	3	4.13544068
E	1	3.947227
E	2	3.99462977
E	3	4.10785979

Analysis of Variance (ANOVA)

- A hypothesis test used to evaluate whether there is a significant difference between **more than 2** groups
  - One-way ANOVA: Comparing groups that differ by one categorical variable
  - Two-way ANOVA: Comparing groups that differ by two categorical variable
  - More variants

- Requirements: residuals are normally distributed, homoscedasticity
- Test Statistic: F-statistic (F-ratio); Test Distribution: F-Distribution
- Non-parametric alternatives: Kruskal-Wallis Test

Solution 2

- To determine if there are significant differences, we can use a two-way ANOVA analysis

	A	B	C	D
1		Strain 1	Strain 2	Strain 3
2	Location A	3.85618697	4.06354438	3.99682508
3	Location B	3.94848585	4.09798208	4.10475134
4	Location C	3.94827211	4.05966656	4.02955828
5	Location D	3.7740048	4.11053965	4.13544068
6	Location E	3.947227	3.99462977	4.10785979
7				

Anova: Two-Factor Without Replication

SUMMARY	*Count*	*Sum*	*Average*	*Variance*
Location A	3	11.9165564	3.97218548	0.01120461
Location B	3	12.1512193	4.05040642	0.00780231
Location C	3	12.037497	4.01249898	0.00332045
Location D	3	12.0199851	4.00666171	0.04075194
Location E	3	12.0497166	4.01657219	0.00681183
Strain 1	5	19.4741767	3.89483535	0.00614312
Strain 2	5	20.3263624	4.06527249	0.00203596
Strain 3	5	20.3744352	4.07488703	0.00344855

ANOVA

urce of Variati	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Rows	0.00932802	4	0.002332	0.50174273	0.73613828	3.83785335
Columns	0.10259978	2	0.05129989	11.0374339	0.00500662	4.45897011
Error	0.03718248	8	0.00464781			
Total	0.14911028	14				

Good luck with content quizzes 6a and 6b!

(Due November 19th at 11:59 PM)